

Lierda NT26-KCN B Series Hardware Design Manual

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Document Revision History

Document Version	Change date	Reviser	Reviewer	Change content
Rev1.0	25-09-28	CWY、ZH	YMX、SLY	Initial version
Rev1.1	25-10-29	CWY、ZH	YMX、SLY	Add descriptions for NT26K2B1 and NT26K2B1-0S modules.
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Rev3.2	26-01-09	CWY、ZH	YMX、SLY	Add descriptions for NT26K0B3-0S and NT26K0B3-M modules.
Rev3.3	26-02-02	CWY、ZH	YMX、SLY	Add descriptions for NT26K2B3 and NT26K2B3-0S modules.

Safety Instructions

Users are responsible for complying with the relevant regulations of other countries regarding wireless communication modules and devices, as well as specific regulations related to the usage environment. By following the safety principles below, personal safety can be ensured, and it helps protect the product and work environment from potential damage. Our company does not assume any related losses caused by customers' failure to comply with these regulations.



Road safety comes first! When you are driving, do not use handheld mobile devices unless they have hands-free functionality. Please stop your vehicle before making a call!



Please turn off mobile terminal devices before boarding. The wireless functions of mobile terminals are prohibited from being turned on during the flight to prevent interference with the aircraft's communication systems. Ignoring this instruction may compromise flight safety and even violate the law.



When in hospitals or healthcare facilities, pay attention to any restrictions on the use of mobile terminal devices. RF interference can cause medical equipment to malfunction, so mobile terminal devices may need to be turned off.



Mobile terminal devices do not guarantee effective connections in all situations, such as when the mobile terminal device has no balance or the SIM is invalid. If you encounter the above situations in an emergency, please remember to use emergency calls, while ensuring that your device is powered on and in an area with sufficient signal strength.



Your mobile terminal device will receive and transmit radio frequency signals when powered on, which can cause radio frequency interference when near televisions, radios, computers, or other electronic devices.



Please keep mobile terminal devices away from flammable gases. When you are near gas stations, oil depots, chemical plants, or explosive operation sites, please turn off mobile terminal devices. Operating electronic devices in any location with potential explosion hazards poses safety risks.

Applicable Module Selection

Serial Number	Module model number	Supported frequency bands	Instructions
1	NT26K0B1	Band1/3/5/8/ 34/38/39/40/41*	VDD_EXT sleep mode power down
2	NT26K2B1	Band1/3/5/8/ 34/38/39/40/41*	VDD_EXT power down in sleep mode, main chip replaced with 716E, lowest power consumption supports HIB mode.
3	NT26K2B1-0S	Band1/3/5/8/ 34/38/39/40/41*	VDD_EXT power-down in sleep mode, main chip replaced with 716E, lowest power consumption supports HIB mode, single card version.
4	NT26K0B1-M	Band3/8/ 34/38/39/40/41*	VDD_EXT power-down in sleep mode, mobile version module
5	NT26K0B3	Band1/3/5/8/ 34/38/39/40/41*	VDD_EXT Sleep Mode Power Down
6	NT26K0B3-0S	Band1/3/5/8/ 34/38/39/40/41*	VDD_EXT power down in sleep mode, single card version
7	NT26K0B3-M	Band3/8/ 34/38/39/40/41*	VDD_EXT power down in sleep mode, mobile version module
8	NT26K2B3	Band1/3/5/8/ 34/38/39/40/41*	The main chip uses EC716E, with VDD_EXT power off in sleep mode.
9	NT26K2B3-0S	Band1/3/5/8/ 34/38/39/40/41*	The main chip uses EC716E, VDD_EXT powers down in sleep mode, single card version.

Precautions

LTE-TDD B41 only supports 140MHz (2535-2675MHz).

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1 Introduction

This document defines the application specifications for the Lierda NT26KxBx series Cat.1 wireless communication modules. It describes their frequency band functions, key features, hardware interfaces, operating modes, electrical characteristics, mechanical specifications, as well as packaging and storage. This can help users quickly grasp the application methods of this Cat.1 module and carry out product design rapidly and flexibly.

1.1 Special characters

Table 1-1 Explanation of Special Symbols

Symbol	Definition
*	Unless otherwise specified, an asterisk (*) marked after module functions, features, interfaces, pin names, AT commands, or parameters indicates that the respective module function, feature, interface, pin name, AT command, or parameter is under development and not currently supported.

2 Product Overview

The NT26KxBx series modules are globally leading Cat.1 wireless communication modules, compliant with the 3GPP R14 Cat.1bis standard. They support bandwidths of 1.4/3/5/10/15/20MHz and feature small size, low power consumption, and strong anti-interference capability. These modules are suitable for various common IoT application scenarios, such as:

- ◆ Smart Meter Reading
- ◆ Intelligent Parking
- ◆ Smart City
- ◆ Intelligent Security
- ◆ Asset Tracking
- ◆ Smart Home Appliances
- ◆ Agriculture and environmental monitoring, etc.

2.1 Key Features

The key features of the NT26KxBx series modules are described as follows:

Table 2-1 Description of Key Module Characteristics

Parameters	Explanation
Module packaging	LCC+LGA
Module size	17.7mm×15.8mm×2.4mm (L×W×H)
Module weight	1.3g
Operating voltage	◆ Power supply voltage: 3.3~4.5V, typical value: 3.8V ◆ Extended supply voltage (1): 3.1~4.5V
VDD_EXT feature	The NT26KxBx series offers two hardware versions. Sleep mode: VDD_EXT power down Sleep mode: VDD_EXT does not power off
Ultra-low power consumption	Shutdown mode: ≤2uA
Transmission power	23dBm±2.7dB (Max)

Antenna interface	50Ω characteristic impedance
USIM	Supported USIM card types: Class B (3.0 V) and Class C (1.8 V), see section 4.3 for details.
UART	1-way main serial port: Used for AT command transmission and data transfer, the supported baud rate defaults to 115200bps. Used for firmware upgrade, the supported baud rate defaults to 921600bps.
	1-way debugging serial port: Used for software debugging and troubleshooting The default baud rate of the serial port is 3 Mbps.
	1-way auxiliary serial port: User-defined function
USB	◆ 1 USB port ◆ Compatible with USB 2.0 (slave mode only), data transfer rate up to 480 Mbps ◆ Used for AT command transmission, data transfer, software debugging, and firmware upgrades
I2C	◆ 1-channel I2C interface ◆ Compliant with I2C bus protocol specifications ◆ Supports standard mode 100KHz, fast mode 400KHz
SPK	◆ 1-channel analog audio interface: ◆ Used for audio, can directly drive low-power headphones or connect to an external PA to drive speakers.
Indicator light	◆ Operating Status Indicator ◆ Network Status Indicator
Communication Interface Characteristics	Supports 3GPP Rel.13/14 Cat.1 radio communication interface and protocols
Network Protocol Features (2)	Supports protocols such as TCP/UDP/HTTP/SSL*/MQTT/FOTA/PPP*/RNDIS*/FTP*/ECM*, etc.
Data Transmission Characteristics	◆ Supports up to Cat.1 FDD and TDD ◆ 1.4/3/5/10/15/20MHz RF bandwidth ◆ Uplink QPSK, 16QAM ◆ Downlink QPSK, 16QAM, 64QAM ◆ LTE-FDD: Maximum downlink speed 10 Mbps, maximum uplink speed 5 Mbps ◆ LTE-TDD: Maximum downlink speed 8.96 Mbps, maximum uplink speed 3.1 Mbps

Firmware Upgrade	◆ Main Serial Port Upgrade ◆ USB Interface Upgrade ◆ DFOTA Upgrade
Operating temperature	◆ Normal operating temperature range (3): -35~75°C ◆ Extended operating temperature range (4): -40~85°C
Storage temperature	Storage temperature range (5): -40~90°C
RoHS	All components comply with EU RoHS standards.

Remarks

(1) For a detailed description of the extended voltage, see section 6.2.

(2) The network protocols supported may vary between different firmware versions; please refer to the documentation of the specific firmware version used.

(3)(4)(5) For specific descriptions regarding temperature, see section 6.6.

2.2 Functional block diagram

The NT26KxBx series modules feature rich peripheral interfaces and RF functions.

The functional block diagram is as follows:

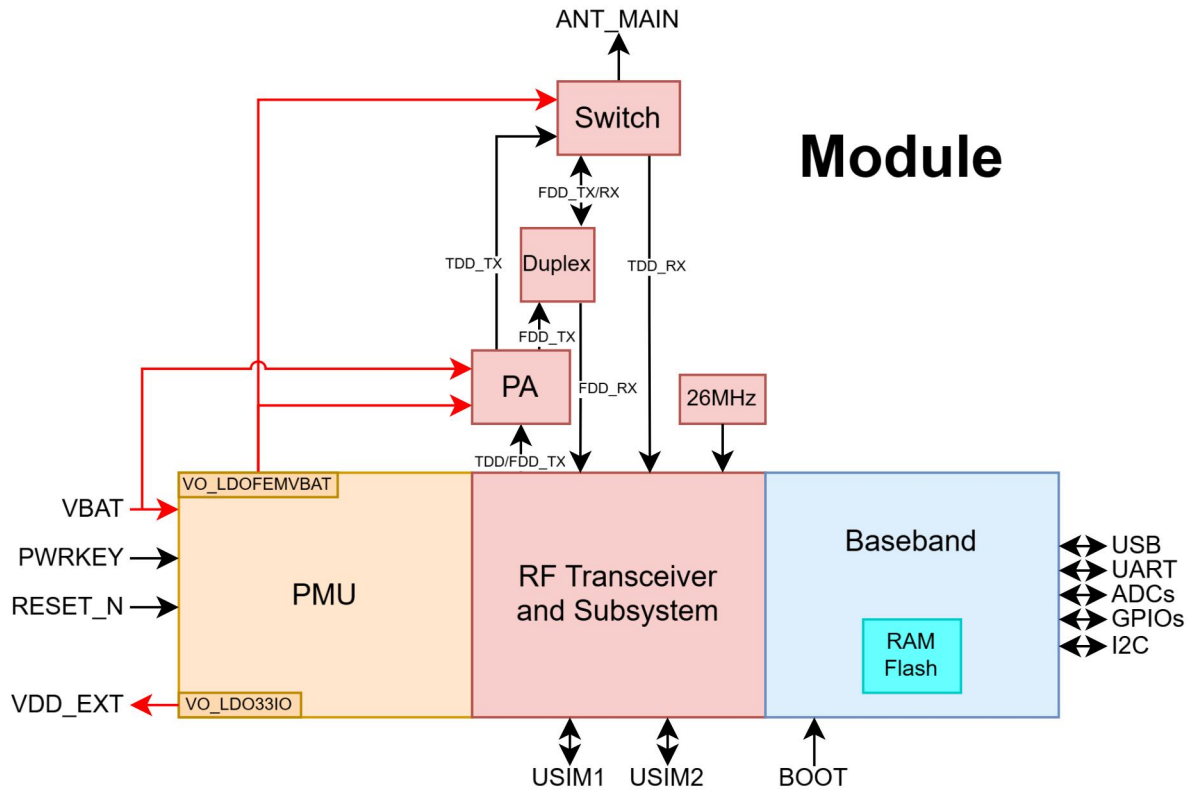


Figure 2.1 Module Functional Block Diagram

2.3 Pinout diagram

The NT26KxBx series module has a total of 109 pins, of which 48 are LCC pins and the remaining 61 are LGA pins. The module pin distribution diagram is as follows:

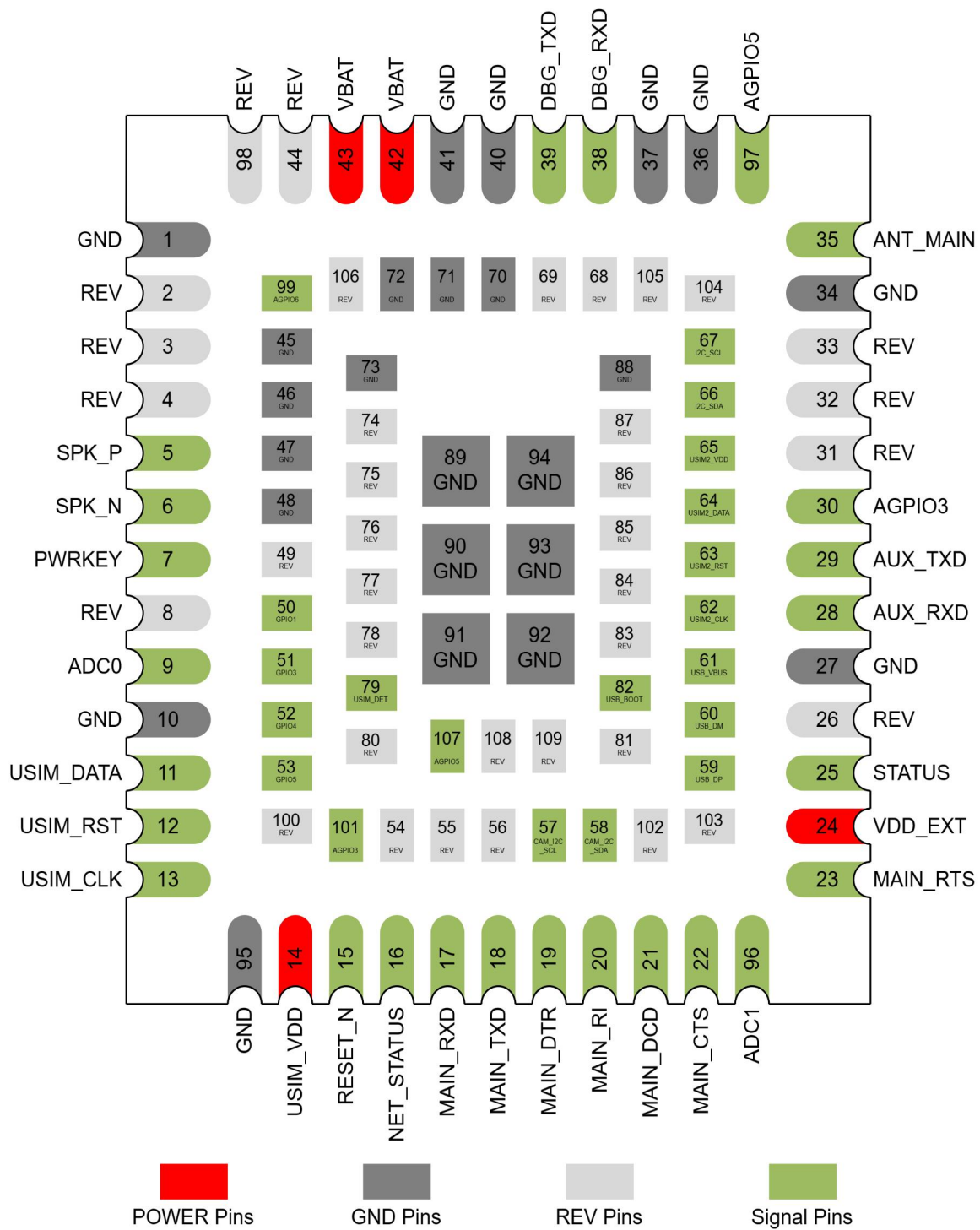


Figure 2.2 Module Pin Assignment Diagram

2.4 Pin Description Table

For better understanding of the application, the following table provides the type definitions of the I/O parameters:

Table 2-2 Type Definition Description of I/O Parameters

I/O Parameter Types	Explanation
DI	Digital input
DO	Digital output
DIO	Digital Input and Output
AI	Simulated input
AO	Simulated output
AIO	Simulated input and output
PI	Power input
PO	Power output
G	Grounding
OD	Drain open circuit

For better understanding of the application, the following table provides the characteristic description of the power domain parameters:

Table 2-3 Module Power Domain Characteristics Description

Power domain parameters Type	DC Characteristics	Instructions	Power supply target
VBAT	Vmax=4.5V Vmin=3.3V Vnorm=3.8V	Module power input, it is recommended to use a 3.8V/1.2A power supply.	Module
VDD_EXT	VILmax=0.2×VDD_EXT VIHmin=0.7×VDD_EXT VOLmax=0.15×VDD_EXT VOHmin=0.8×VDD_EXT	Default 1.8V; Different hardware versions: In sleep mode, the VDD_EXT power-off version has a driving capability of 120mA. In sleep mode, the VDD_EXT non-power-off version has a driving capability of 3mA.	VDD_EXT
VO_LDOIO	VOLmax=0.2×VO_LDOIO VOHmin=0.7×VO_LDOIO	Default 1.8V, configurable via software;	UART GPIO

	VILmax=0.15×VO_LDOIO VIHmin=0.8×VO_LDOIO	Power loss in sleep mode	
VDD18AON	VILmax=0.2×VDD18AON VIHmin=0.7×VDD18AON VOLmax=0.15×VDD18AON VOHmin=0.8×VDD18AON	No power loss in sleep mode	WAKEUP RESET
LDO_AONIO	VILmax=0.2×LDO_AONIO VIHmin=0.7×LDO_AONIO VOLmax=0.15×LDO_AONIO VOHmin=0.8×LDO_AONIO	Default 1.8V, configurable via software; No power drain in sleep mode	AGPIO
VO_LDOSIM	Vnorm=1.8/3.0V VOLmax=0.15×VO_LDOSIM VOHmin=0.8×VO_LDOSIM VILmax=0.2×VO_LDOSIM VIHmin=0.7×VO_LDOSIM	Dedicated power supply for SIM cards, supporting 1.8/3.0V cards	USIM

Remarks

- ◆ Power domain: An area within a chip or system that is divided for independent power supply, where each domain can individually control power on/off, voltage levels, or power consumption modes;
- ◆ VDD_EXT and VO_LDOIO are the same power supply.

Table 2-4 Module Pin Description

1. Power					
Pin number	Pin name	I/O	Description	DC Characteristics	Remarks
42,43	VBAT	PI	Power supply	Vmax=4.5V Vmin=3.3V Vnorm=3.8V	The external power supply needs to provide a current carrying capacity of more than 1.2A.
24	VDD_EXT	PO	Output power	Vnorm=1.8V IOmax=120mA/3mA	Can be used for external circuit pull-up or reference level; in sleep mode, there are two modes: power-down and no power-down; if not

					used, leave it floating, and it is recommended to reserve a test point.
1,10,27,34, 36~37,40~41,45 ~48,70~72,73,88 ~95	GND	G	Grounding	-	-

2. Power On/Off and Reset

Pin number	Pin name	I/O	Description	DC Characteristics	Remarks
7	PWRKEY	DI	Module power on/off pin	$V_{ILmax}=0.5V$	Internal pull-up, active low.
15	RESET_N	DI	Module reset pin	$V_{ILmax}=0.5V$	Internal pull-up, active low.

3、USB

Pin number	Pin name	I/O	Description	DC Characteristics	Remarks
59	USB_DP*	DIO	USB differential data (+)	-	Complies with USB 2.0 specifications. Requires 90 Ω differential impedance. Test points need to be reserved. Leave them floating if not used. (Disabled by default)
60	USB_DM*	DIO	USB differential data (-)	-	
61	USB_VBUS*	DI	USB_VBUS input wake-up	$V_{max} = 5.25 V$ $V_{min} = 3.0 V$ $V_{nom} = 5.0 V$	

4. Serial Port UART

Pin number	Pin name	I/O	Description	DC characteristics	Remarks
17	MAIN_RXD	DI	Main serial port receiving data	VO_LDOIO	If not used, leave floating

18	MAIN_TXD	DO	Main serial port data transmission	VO_LDOIO	If not used, leave floating
19	MAIN_DTR	DI	Main serial port data terminal ready	VDD18AON	AGPIO function, power domain LDO_AONIO, does not lose power during sleep. WAKEUP function, power domain VDD18AON, does not lose power during sleep.
20	MAIN_RI	DO	Main serial port outputs ring indication	LDO_AONIO/VDD18AON	Default high level, when the module has a URC to report, output a low level for 120ms, and restore to high level after data output is completed.
21	MAIN_DCD*	DO	Main serial port output carrier detection	VO_LDOIO	Internally connected in parallel with pin 66, cannot be used simultaneously.
22	MAIN_CTS*	DO	Main serial port DTE clear send	VO_LDOIO	Internally connected in parallel with pin 57, cannot be used simultaneously.
23	MAIN_RTS*	DI	Main serial port DTE requests to	VO_LDOIO	Internally connected in parallel with pin

			send		58, cannot be used simultaneously.
38	DBG_RXD	DI	Debugging serial port data reception	VO_LDOIO	If not used, leave floating
39	DBG_TXD	DO	Debug serial port data transmission	VO_LDOIO	If not used, leave floating
28	AUX_RXD	DI	Assist with serial port data reception	VO_LDOIO	Internally connected in parallel with pin 52, cannot be used simultaneously.
29	AUX_TXD	DO	Assist in sending data via serial port	VO_LDOIO	Internally connected in parallel with pin 53, cannot be used simultaneously.

5. External USIM Card Interface

Pin number	Pin name	I/O	Description	DC Characteristics	Remarks
11	USIM_DATA	DIO	SIM card data cable	VO_LDOSIM	
12	USIM_RST	DIO	SIM card reset line	VO_LDOSIM	
13	USIM_CLK	DIO	SIM card clock line	VO_LDOSIM	
14	USIM_VDD	PO	SIM card power supply	VO_LDOSIM	1.8/3.0V
62	USIM2_CLK	DIO	SIM2 (secondary card) clock line	VO_LDOIO	
63	USIM2_RST	DIO	SIM2 (secondary card) reset line	VO_LDOIO	Internally connected in parallel with pin 50, cannot be used

					simultaneously.
64	USIM2_DATA	DIO	SIM2 (Secondary SIM) Data Cable	VO_LDOIO	Internally connected in parallel with pin 51, cannot be used simultaneously.
65	USIM2_VDD	PO	SIM2 (Secondary SIM) Power Supply	VO_LDOSIM	The same power supply as USIM_VDD
79	USIM_DET	DI	SIM Card Status Signal Detection	VDD18AON	Active high

6. Antenna Interface RF

Pin number	Pin Name	I/O	Description	DC characteristics	Remarks
35	ANT_MAIN	AIO	RF antenna interface	-	50Ω characteristic impedance

7. I2C Interface

Pin number	Pin name	I/O	Description	DC Characteristics	Remarks
57	CAM_I2C_SCL	DO	I2C Serial Clock	VO_LDOIO	Internally connected in parallel with pin 22, cannot be used simultaneously.
58	CAM_I2C_SDA	DIO	I2C serial data	VO_LDOIO	Internally connected in parallel with pin 23, cannot be used simultaneously.
66	I2C_SDA	DIO	I2C serial data	VO_LDOIO	Internally connected in parallel with pin

					21, cannot be used simultaneously; when used as an I2C data signal, an external 4.7K pull-up resistor is required, otherwise leave it floating.
67	I2C_SCL	DO	I2C serial clock	VO_LDOIO	The I2C clock signal requires an external 4.7K pull-up resistor; if not used, it will be left floating.

8. Simulated audio interface*

Pin number	Pin name	I/O	Description	DC characteristics	Remarks
5	SPK_P	DO	Audio true differential output	VO_LDOIO	Internally connected in parallel with pin 82, cannot be used simultaneously.
6	SPK_N	DO	Audio negative differential output	Internal grounding	If not used, leave floating

9. ADC Interface

Pin number	Pin name	I/O	Description	DC Characteristics	Remarks
9	ADC0	AI	General ADC Interface	Voltage range: 0~1.05V	If not used, leave floating
96	ADC1	AI	General ADC Interface		If not used, leave floating

10. Status Indicator					
Pin number	Pin name	I/O	Description	DC Characteristics	Remarks
16	NET_STATUS	DO	Network Status Indicator	VDD18AON	If not used, leave floating
25	STATUS	DO	Operating Status Indicator	LDO_AONIO/VDD18AON	AGPIO function, power domain LDO_AONIO, does not lose power during sleep. WAKEUP function, power domain VDD18AON, does not lose power during sleep.

11. Other interfaces					
Pin number	Pin name	I/O	Description	DC Characteristics	Remarks
82	USB_BOOT	DI	Emergency download mode control. Before the module powers on, pull up to VDD_EXT. After the module starts, it will enter emergency download mode.	VO_LDOIO	Internally connected in parallel with pin 5, cannot be used simultaneously; The module has a default weak internal pull-down, active high. Before the module powers on normally, do not pull this pin high. It is recommended to reserve test

					points.
50	GPIO1	DIO	General GPIO	VO_LDOIO	Internally connected in parallel with pin 63, cannot be used simultaneously.
51	GPIO3	DIO	General GPIO	VO_LDOIO	Internally connected in parallel with pin 64, cannot be used simultaneously.
52	GPIO4	DIO	General GPIO	VO_LDOIO	Internally connected in parallel with pin 28, cannot be used simultaneously.
53	GPIO5	DIO	General GPIO	VO_LDOIO	Internally connected in parallel with pin 29, cannot be used simultaneously.
97,107	AGPIO5	DIO	General-purpose GPIO, can be maintained in sleep mode	LDO_AONIO	If not used, leave floating
99	AGPIO6	DIO	General GPIO, can be maintained in sleep mode	LDO_AONIO	Currently used as the internal pull-up power for pin 62, external use is prohibited.
30,101	AGPIO3	DIO	General-purpose GPIO, can be maintained in sleep mode	LDO_AONIO	If not used, leave floating

12. Reserved interface					
Pin number	Pin Name	I/O	Description	DC characteristics	Remarks
2~4,8,26,31~33,44,49,54~56,74,74~78,80~81,83~87,98,100,102~105,106,108~109	REV	-	-	-	Undefined pins are recommended to be left floating and are not recommended for customer use (pins 68, 69, 75, 86, 87, and 105 are prohibited from use and should be left floating).

Note: "*" indicates that the feature is under development.

Remarks

1.The AGPIO driving capability is limited, and the high-level output is prone to fluctuation. It is only suitable for signal level recognition and cannot be used as a power supply for peripherals.

2.2. It is recommended to leave pins PIN8, PIN68, PIN69, PIN75, PIN86, PIN87, and PIN105 floating; grounding is prohibited.

3.3. AGPIOWU drive capability is less than 30uA, and it can only be used as an input, not as an output.

4. It is not recommended to use PIN50 and PIN51; it is advised to leave them floating.

2.5 Evaluation Kit

Lierda can provide complete evaluation and development kits, including module adapter boards, Cat.1 & NB-IoT module universal development boards, etc. Please feel free to contact us for inquiries.

3 Work Characteristics

3.1 Work Mode

The NT26KxBx series modules have multiple operating modes. The table below briefly describes several common operating modes of the modules:

Table 3-1 Working Modes

Mode	Status Description
Normal working mode	Data Transmission: Network connection is normal. Module power consumption depends on the network environment and data transmission rate.
	Idle state: The software is running normally. The module is connected to the network but does not exchange data with the network.
Minimum Function Mode	AT+CFUN=0 can set the module to minimum functionality mode, where both the RF and USIM card do not operate.
Sleep mode	The module's power consumption will be reduced to an extremely low level, but the module can still receive paging, short messages, and TCP/UDP data.
Shutdown mode	VBAT power supply is not interrupted, software stops working.

Remarks

1. For AT+CFUN related commands, please refer to the "AT Command Manual".
2. For detailed working modes and corresponding power consumption data, please refer to the PMU application documentation.

3.2 Sleep mode

The module can enter sleep mode through AT commands. In sleep mode, the module can reduce power consumption to a very low level, while the module's RF can still receive network information.

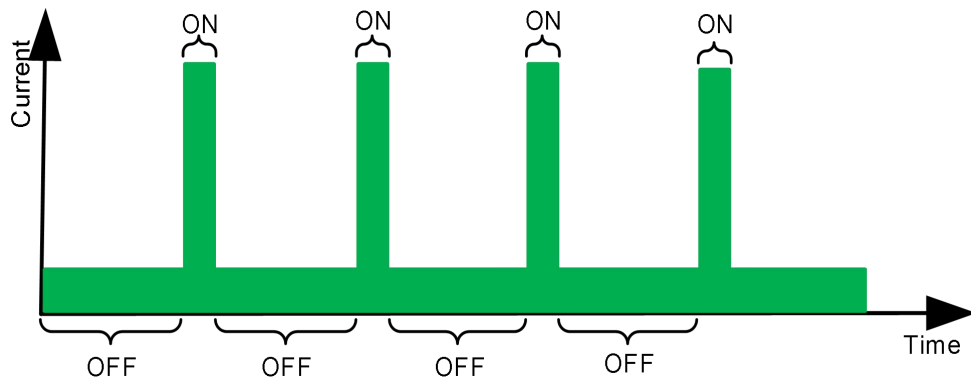


Figure 3.1 Power Consumption States of the Module in Sleep Mode

In sleep mode, the module's peripherals are powered down to reduce power consumption, but the RF remains in intermittent reception mode and can still receive network information. When the module has a URC to report, it will wake up the host through the MAIN_RI pin.

3.2.1 Main Serial Port Application Scenarios

If the module and the host communicate via UART, the following two conditions must be met simultaneously for the module to enter sleep mode:

- ◆ Execute the AT+QSCLK=1 or AT+LSCLK=1 command (refer to the "AT Command Manual" for specific commands) to enable the sleep function;
- ◆ Ensure that MAIN_DTR remains at a high level or is left floating.

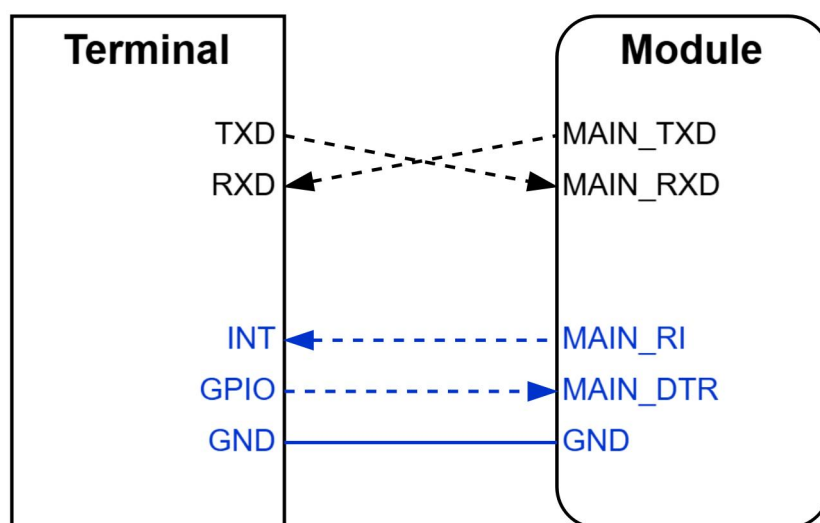


Figure 3.2 Application Circuit of Sleep Mode UART

- ◆ When the module has a URC to report, the module notifies the host through

MAIN_RI.

- ◆ The host pulling MAIN_DTR low can wake up the module.
- ◆ If the host continuously pulls MAIN_DTR low, the module can be prevented from entering sleep mode.
- ◆ If the sleep function is enabled using AT+QSCLK=2 or AT+LSCLK=2 (please refer to the "AT Command Manual" for specific commands), the module can be awakened by sending any command through the main UART interface.

3.2.2 USB Application Scenarios*

In USB application scenarios, the module enters sleep mode only when the following three conditions are met simultaneously:

- ◆ Execute AT+QSCLK=1/AT+LSCLK=1 (choose according to the "AT Command Manual");
- ◆ Ensure MAIN_DTR remains at a high level or is left floating;
- ◆ The host USB bus connected to the module's USB interface enters a suspended state.

3.2.2.1 Supports USB remote wake-up function*

The host supports USB suspend and wake-up as well as USB remote wake-up functions.

Please refer to the following diagram for the connection between the module and the host:

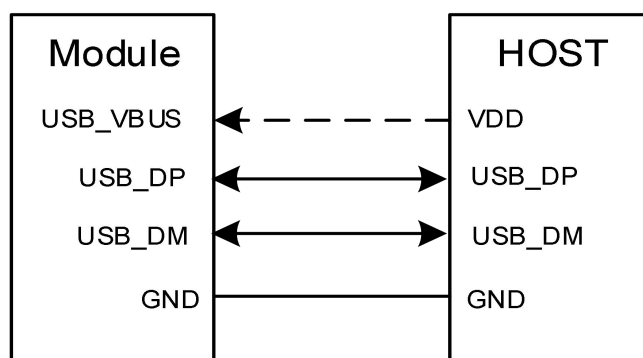


Figure 3.3 Sleep Application with USB Remote Wake-up Function

- ◆ Sending data to the module via USB will wake up the module.
- ◆ When the module has a URC report, it will send a remote wake-up signal through the USB bus to wake up the host.

3.2.2.2 Supports USB suspend and wake-up as well as MAIN_RI function*

If the host supports USB suspend and wake-up but does not support USB remote wake-up functionality, the host needs to be awakened by the module's MAIN_RI signal.

The connection between the module and the host refers to the following diagram:

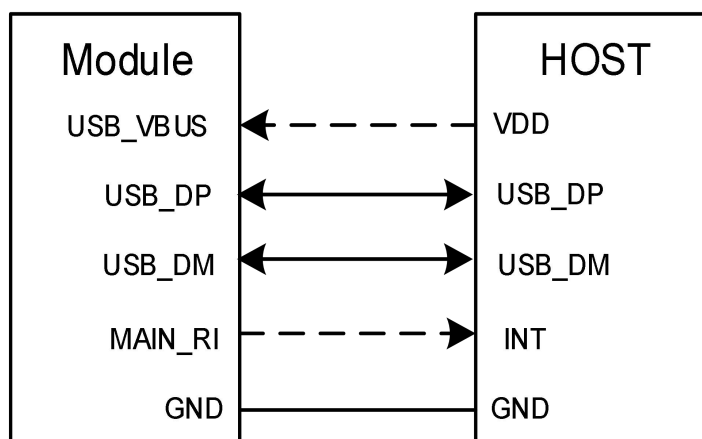


Figure 3.4 Sleep Application with MAIN_RI Function

- ◆ Sending data to the module via USB will wake up the module;
- ◆ When the module has a URC report, the module can wake up the host through

MAIN_RI.

3.2.2.3 USB suspend function not supported*

When the host does not support USB suspend functionality, the following three conditions must be met simultaneously for the module to enter sleep mode:

- ◆ Execute AT+QSClk=1 or AT+LSCLK=1 (for specific commands, please refer to the "AT Command Manual");
- ◆ Ensure MAIN_DTR remains at a high level or is left floating;
- ◆ Disconnect USB_VBUS power supply.

If the host does not support the USB suspend function, the module can enter sleep mode by disconnecting USB_VBUS through an external control circuit.

The connection between the module and the host is shown in the diagram below:

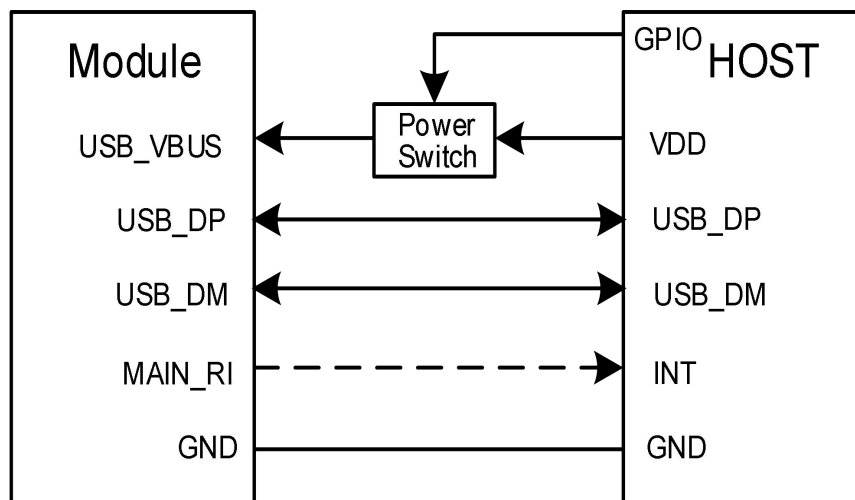


Figure 3.5 Sleep applications that do not support USB suspend function

Remarks

Restoring USB_VBUS power supply can wake up the module.

3.3 Airplane mode

When the module enters flight mode, the RF function is disabled, and all AT commands

related to RF are inaccessible. The module can enter flight mode through the following methods:

It can be set by sending the AT+CFUN=<fun> command. The <fun> parameter can be set to 0 or 1.

- ◆ AT+CFUN=0: Minimum functionality mode (turns off RF functions and USIM card).
- ◆ AT+CFUN=1: Full functionality mode (default).

3.4 Power Supply Design

3.4.1 Power Supply Introduction

The NT26KxBx series modules provide 2 VBAT pins for connecting external power supplies. Additionally, VDD_EXT is the module's external output power supply and can be used as a reference voltage. The interface description is as follows:

Table 3-2 Power Pin Definitions

Pin number	Pin name	Description	Minimum value	Typical value	Maximum value	Unit
42,43	VBAT	Modular power supply	3.3	3.8	4.5	V
24	VDD_EXT	Output power	-	1.8	-	V

The module's VBAT power supply range is 3.3 to 4.5V. It is necessary to ensure that the voltage drop caused by the module during operation does not cause the minimum input voltage to fall below 3.3V. The diagram below illustrates the VBAT voltage drop of the module during burst transmission:

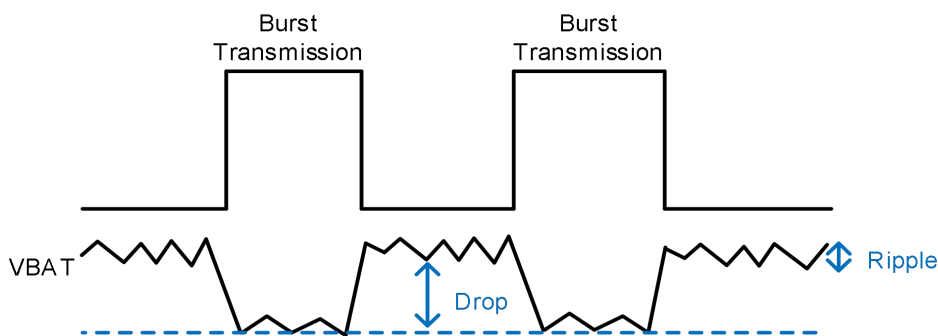


Figure 3.6 Burst Transmission Power Supply Requirements

Remarks

In sleep mode, VDD_EXT will be powered off.

3.4.2 Power Supply Design

When the module operates at the maximum transmission power of the 4G network, the transient operating current in the live network can reach 1.2A, which may cause a drop in the power supply voltage. Under any circumstances, it is necessary to ensure that the module power supply voltage remains above 3.3V; otherwise, the module may experience unexpected situations such as rebooting. To reduce voltage drops, it is recommended to use a 100uF low ESR ($ESR < 0.7\Omega$) filter capacitor. It is also recommended to reserve three chip-type multilayer ceramic capacitors (MLCCs) with good ESR performance for VBAT, and the capacitors should be placed close to the VBAT pin. Additionally, it is suggested to add a high-power TVS diode with low clamping voltage and high reverse pulse current (Ipp) at the VBAT terminal to enhance the module's surge voltage withstand capability.

The power supply reference circuit is as follows:

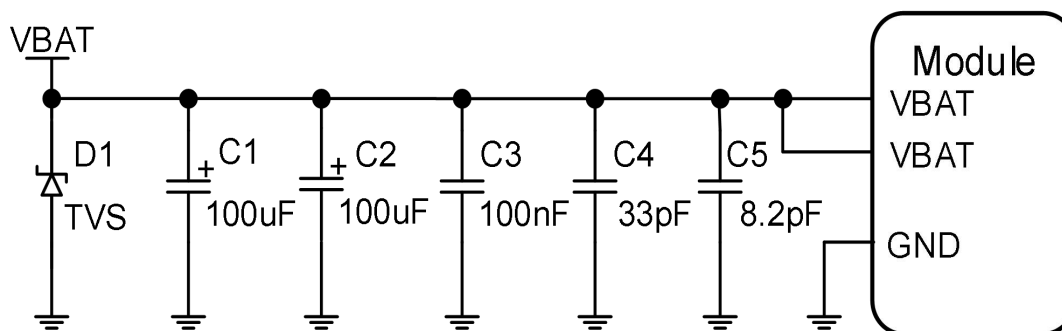


Figure 3.7 Module Power Supply Circuit Reference

In PCB design, the longer the VBAT trace, the wider the trace should be. It is recommended that the trace width be no less than 2mm. The GND plane in the power section should be as complete as possible with multiple ground vias. At the same time, decoupling capacitors should be placed as close as possible to the VBAT pin of the module. Among them:

- ◆ D1 is a high-power TVS diode with a low clamping voltage;
- ◆ C1 and C2 are 100uF low ESR tantalum capacitors to enhance the power supply's current sustain capability and stabilize the voltage;
- ◆ C3~C5 are ceramic package filter capacitors of 100nF, 33pF, and 8.2pF respectively, used to eliminate high-frequency interference.

3.5 Power on

3.5.1 Startup Introduction

The module can be powered on through the PWRKEY pin, with the pin description as follows:

Table 3-3 PWRKEY Pin Description

Pin number	Pin Name	I/O	Description	Remarks
7	PWRKEY	DI	Module power-on, active low.	Default is high level

The control logic of PWRKEY is as follows:

- ◆ When the module is powered off, it can be turned on by pulling the PWRKEY low for at least 1 second;

3.5.2 Power-On Circuit Reference Design

It is recommended to use an open-collector drive circuit or directly control the power-on through a button. To prevent static discharge caused by contact, a TVS should be placed near the button for ESD protection. The reference circuit is as follows:

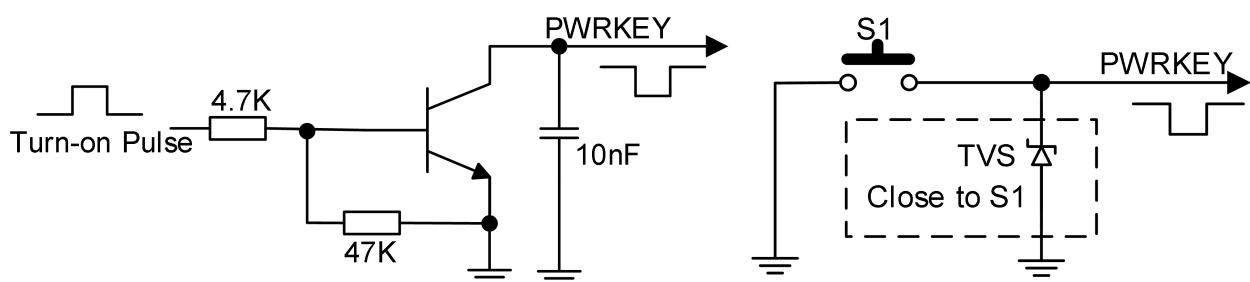


Figure 3.8 Power On/Off Circuit

3.5.3 Power-on sequence

The power-on sequence of the module is shown in the following diagram:

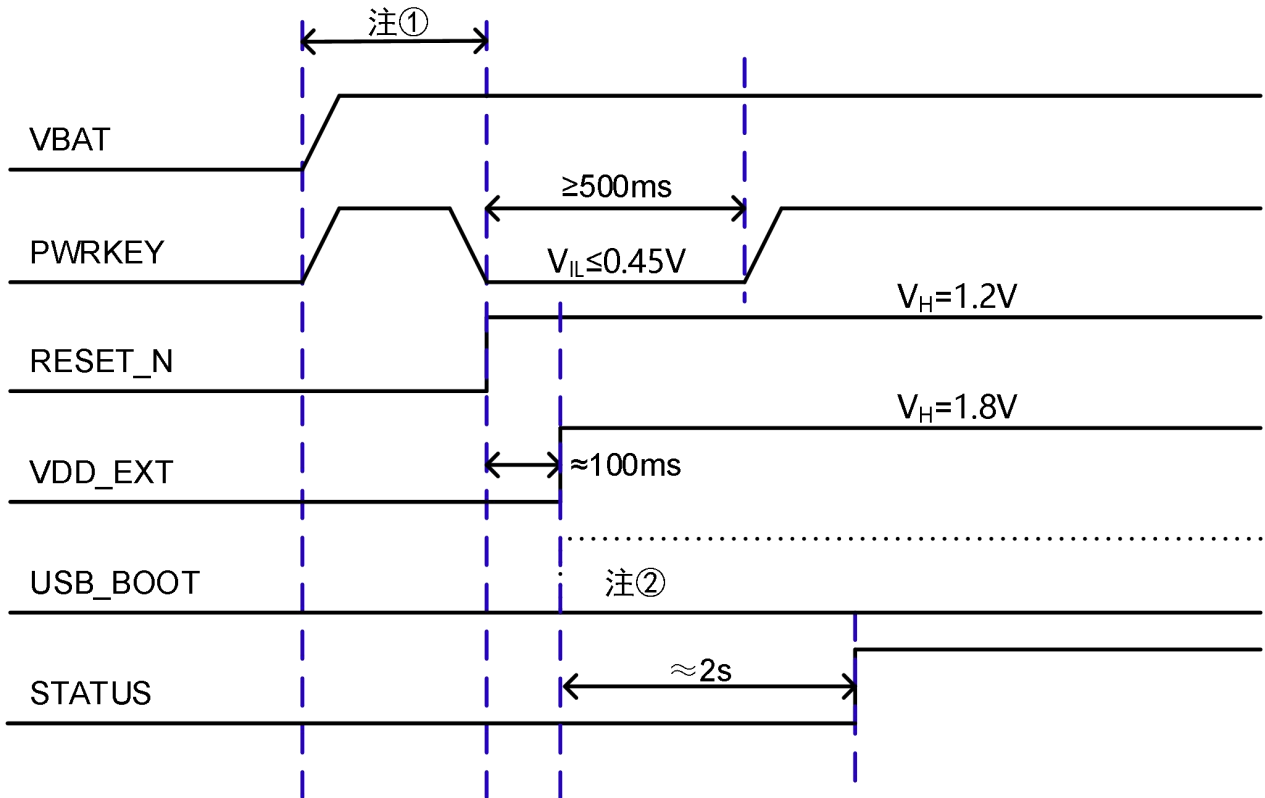


Figure 3.9 Power-On Startup Timing Diagram

Precautions

◆ The initial state of VBAT power-on must be less than 0.1V, and the time for VBAT to rise from 0V to 2.5V should be less than 10ms; before pulling down the PWRKEY pin, ensure the VBAT voltage is stable. It is recommended that the time interval between VBAT power-on and pulling down the PWRKEY pin is no less than 30ms.

◆ The USB_BOOT pin should be pulled to VDD_EXT before the module powers on; when the module powers on, it will enter emergency download mode.

◆ If automatic power-on is required without the need for a shutdown function, the PWRKEY can be directly pulled down to ground. It is recommended to use a 4.7k Ω

pull-down resistor, or control the PWRKEY to a low level via GPIO before the module powers on and keep it low continuously.

◆ The module's main serial port sends the characters "Lierda" or "RDY" to indicate that the startup is complete. Please refer to the "AT Command Manual" for details.

3.6 Shutdown

The module can be properly shut down through the following methods:

- ◆ Control the module shutdown through the PWRKEY pin.
- ◆ Send the AT+QPOWD or AT+LPOWD (please refer to the "AT Command Manual" for specific commands) command to power off.

3.6.1 PWRKEY shutdown

In the power-on state, pull PWRKEY low for 650ms continuously and then release it. The module will execute the shutdown process, with the timing sequence as follows:

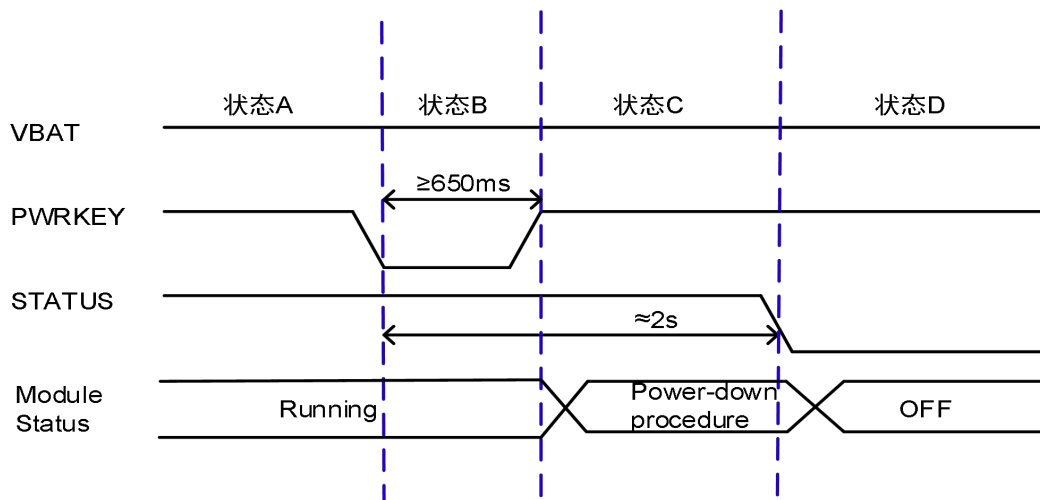


Figure 3.10 PWRKEY Shutdown Timing Diagram

PWRKEY Shutdown Timing Description:

- ◆ Status A: Module operating status;
- ◆ State B: Lowering the PERKEY state, duration at least 650ms or more;
- ◆ State C: The module performs a system shutdown internally, with the time from pulling PWRKEY low to shutdown completion being approximately 2 seconds;

- ◆ Status D: The module has entered shutdown mode.

3.6.2 AT command shutdown

Executing AT+QPOWD or AT+LPOWD (please refer to the "AT Command Manual" for specific commands) can shut down the module. This operation has the same timing and effect as pulling PWRKEY low to power off.

Precautions

- ◆ When the module is operating normally, do not immediately cut off the module's power, as this may damage the internal Flash data of the module. It is recommended to first shut down the module using the PWRKEY or AT command before disconnecting the power.
- ◆ When using AT commands to shut down, please ensure that the PWRKEY remains at a high level after the shutdown command is executed; otherwise, the device may restart after shutdown, and disconnecting the power at this time risks damaging the Flash.
- ◆ If powering on by keeping PWRKEY commonly grounded, it is recommended to use AT+QPOWD/AT+LPOWD (according to the "AT Command Manual") to perform the shutdown process before disconnecting the power.

3.7 Reset

3.7.1 Reset Introduction

The RESET_N pin can be used for module reset. Pulling the RESET_N pin low for at least 300ms and then releasing it will reset the module. The description of the RESET_N pin is as follows:

Table 3-4 Reset Pin Description

Pin number	Pin name	I/O	Description	DC Characteristics	Remarks
15	RESET_N	DI	Module reset	VILmax=0.5V	There is a weak pull-up inside, active low, leave floating if not used.

3.7.2 Reset Circuit Reference Design

The RESET_N signal is relatively sensitive to interference. It is recommended that the module interface wiring be kept as short as possible and should be grounded. The typical reference hardware circuit application is as follows; two circuit diagrams can be selected based on actual requirements:

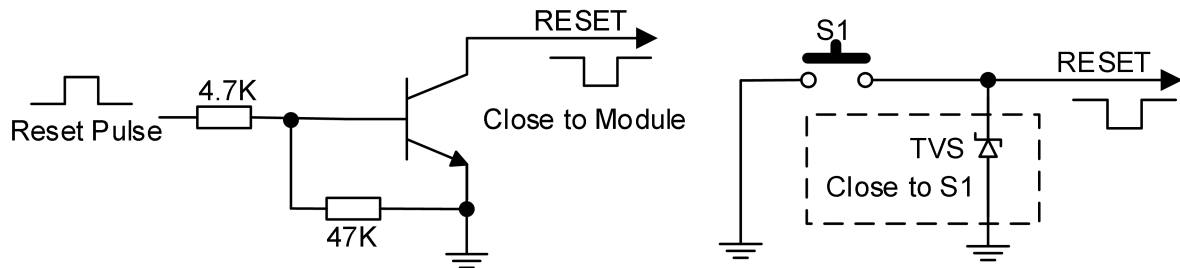


Figure 3.11 Reset Circuit Reference Design

3.7.3 Reset timing

The reset timing sequence of the module is as follows:

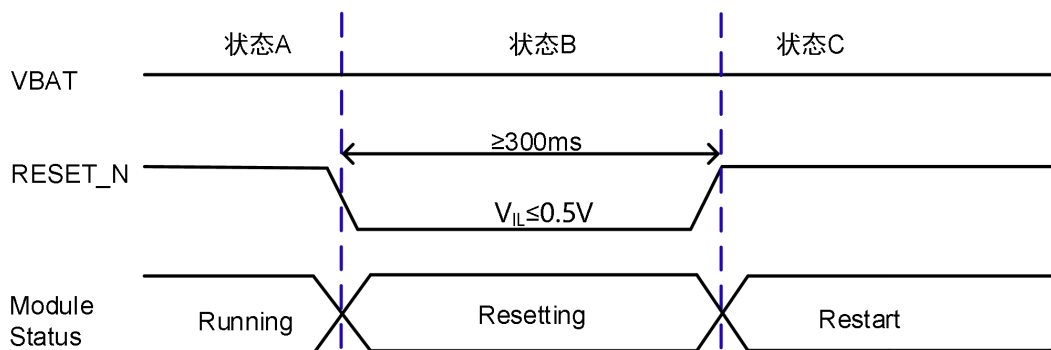


Figure 3.12 Reset Timing Diagram

Reset timing description:

- ◆ Status A: Module operating normally;
- ◆ State B: Pull the RESET_N line low for at least 300ms;
- ◆ Status C: The module begins to reboot.

4 Application Interface

The NT26KxBx series modules offer a wide range of peripheral functions, mainly including the following application interfaces:

- ◆ 1*USB 2.0 interface*
- ◆ 3*UART interfaces
- ◆ 2*USIM interfaces
- ◆ 1*I2C interface
- ◆ 1*Simulated audio interface

4.1 USB interface*

The NT26KxBx series modules support USB 2.0 interface, comply with USB 2.0 specifications, and support full-speed (12 Mbps) and high-speed (480 Mbps) modes. The modules only support USB slave mode. This interface can be used for AT command sending, data transmission, software debugging, and firmware upgrading. The interface definition is as follows:

Table 4-1 USB Interface Pin Description

Pin number	Pin name	I/O	Description	DC characteristics	Remarks
59	USB_DP	DIO	USB differential data (+)	-	Complies with USB 2.0 specifications. Requires 90 Ω differential impedance. Test points must be reserved. If not used, leave them floating.
60	USB_DM	DIO	USB differential data (-)	-	Complies with USB 2.0 specifications. Requires 90 Ω

					differential impedance. Test points must be reserved. If not used, leave them floating.
61	USB_VBUS	DI	USB plug-in wake-up module	Vmax = 5.25 V Vmin = 3.0 V	In sleep mode, inserting a USB will wake up the module.
82	USB_BOOT	DI	USB Mode Configuration Interface	VO_LDOIO	Internal weak pull-down, active high. Do not pull up this pin before the module powers on normally. It is recommended to reserve a test point; leave it floating if not used.

4.1.1 USB Circuit Reference Design

The module USB interface can connect to a USB connector or the MCU's USB interface. The reference schematic for connecting to the USB connector is as follows:

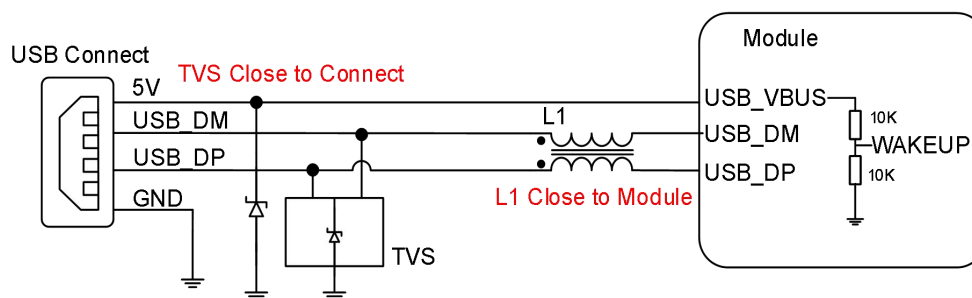


Figure 4.1 USB Interface Reference Design

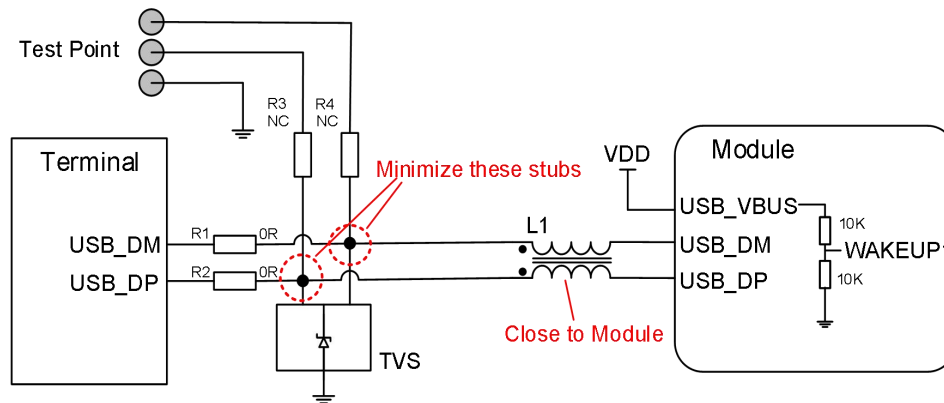


Figure 4.2 USB Connection MCU Reference Design

Precautions

◆ To ensure USB performance, the USB_DP and USB_DM lines should maintain a 90 Ω differential impedance, with grounding around them. It is recommended to place a common mode choke L1 in series between the MCU and the module to prevent EMI interference from the USB signal. L1 should be placed close to the module.

◆ USB wiring should be routed away from crystal oscillators, oscillators, magnetic devices, RF signals, and strong signal areas. It is recommended to use inner layer differential routing with three-dimensional grounding on all sides (top, bottom, left, and right).

◆ If the module USB interface and MCU are connected through a connector, a TVS diode protection should be added at the interface. When selecting ESD protection devices, special attention should be paid to ensure that their parasitic capacitance does not exceed 2pF and they should be placed as close to the USB interface as possible.

◆ Reserve resistors R1~R4; normally, R1~R2 are soldered and R3~R4 are not, allowing normal connection to the MCU. For debugging, R3~R4 can be soldered and R1~R2 unsoldered to switch to the test points. To meet USB data line signal integrity requirements, the branching wires connecting to the test points should be kept as short as possible.

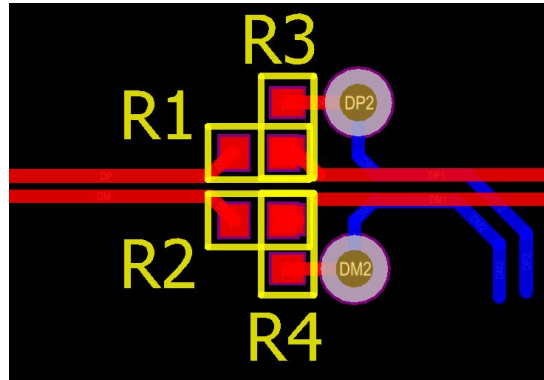


Figure 4.3 USB Jumper Selection Wiring Diagram

4.1.2 USB data transmission

When the module powers on or resets, if USB_BOOT is detected to be at a low level, the USB port will enumerate two serial ports for AT command transmission and log output.

4.1.3 USB firmware download

When the module is powered on or reset/restarted, if USB_BOOT is detected to be at a high level, the USB port will enumerate a dedicated download serial port. At this time, the download tool can upgrade the firmware via the USB interface. If no download tool is connected within 15 seconds, the module will re-enumerate two serial ports according to the default USB configuration.

The USB_BOOT interface reference design is as follows:

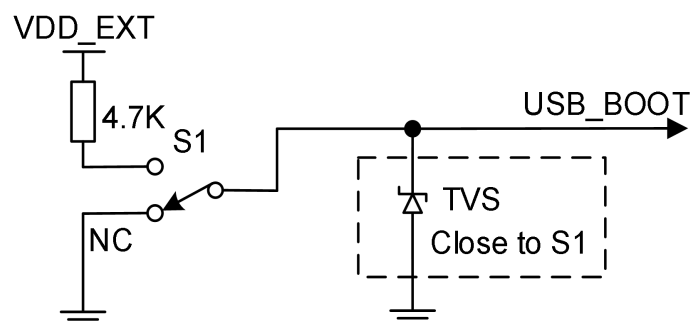


Figure 4.4 USB_BOOT Reference Design

4.2 UART Communication

The module provides three general asynchronous transceivers: the main serial port, the debug serial port, and the auxiliary serial port. The baud rate of the main serial port is configurable, while the debug serial port is only used for debugging and testing purposes. The serial port pin definitions are as follows:

Table 4-2 Serial Port Pin Definitions

Interface	Pin number	Pin name	Description	Remarks
Main serial port	17	MAIN_RXD	Main serial port receiving data	Non-sleep state: VDD_EXT power domain; Sleep state: Switch to internal AON power domain
	18	MAIN_TXD	Main serial port data transmission	If not used, leave floating
	19	MAIN_DTR	Main serial port data terminal ready	In sleep mode, it can serve as a wake-up pin.
	20	MAIN_RI	Main serial port outputs ring indication	If not used, leave floating
	21	MAIN_DCD*	Main serial port output carrier detection	Not supported yet
	22	MAIN_CTS*	Main serial port DTE clear send	Internally connected in parallel with pin 57, cannot be used simultaneously.
	23	MAIN_RTS*	Main serial port DTE requests to send	Internally connected in parallel with pin 58, cannot be used simultaneously.
Debugging serial port	38	DBG_RXD	Debugging serial port data reception	If not used, leave floating
	39	DBG_TXD	Debug serial port data	If not used, leave floating

			transmission	
Auxiliary serial port	28	AUX_RXD*	Assist with serial port data reception	Not supported yet
	29	AUX_TXD*	Assist with serial port data transmission	Not supported yet

Precautions

- ◆ The serial port interface belongs to the VO_LDOIO power domain, with a default voltage level of 1.8V;
- ◆ The VO_LDOIO power domain will lose power in sleep mode, and MAIN_RXD switches to the internal 1.2V power supply;
- ◆ The MAIN_RXD pin has an internal pull-up resistor, so no external pull-up resistor is needed, and it has reverse current protection.
- ◆ Pay attention to the consistency of voltage levels when using the serial port; otherwise, leakage current may easily occur.

4.2.1 Serial port application

Main Features of the Primary Serial Port:

- ◆ Used for AT command communication and data transmission, with a default baud rate of 115200bps (8-N-1);
- ◆ Configurable to: 4800/9600/19200/38400/57600/115200/230400/460800/921600 bps;
- ◆ Used for firmware upgrades, the default baud rate during the upgrade is 921600bps;
- ◆ Supports low-power wake-up module functionality.

Characteristics of debugging serial ports:

- ◆ Through the dedicated tools provided by the platform, log information can be viewed for software debugging, with a default rate of 3000000bps.

The schematic diagram of the serial port connection method is as follows:

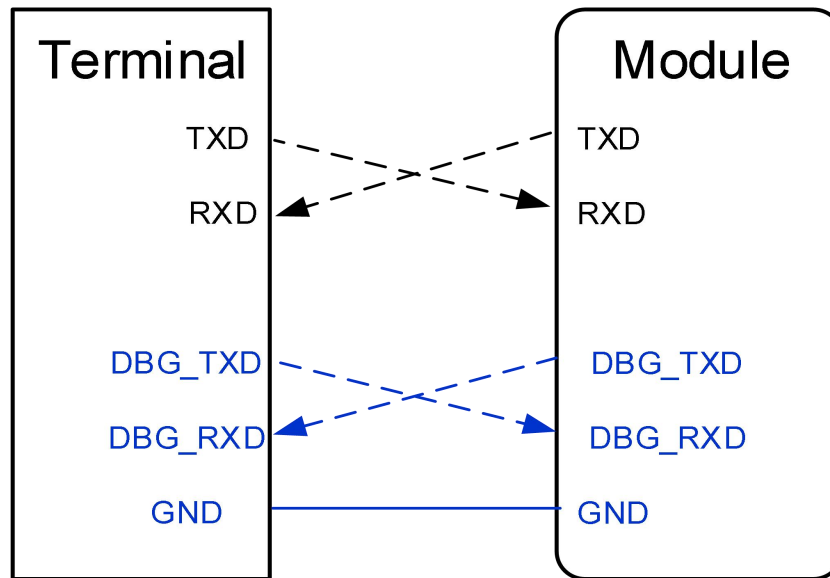


Figure 4.5 Serial Port Connection Diagram

4.2.2 Serial Port Circuit Reference Design

The main factors to consider for a suitable serial port level conversion circuit include whether it meets the operating speed of the serial port, whether there are low power consumption requirements, and whether its power consumption meets the requirements. It is recommended to refer to the following level matching circuit:

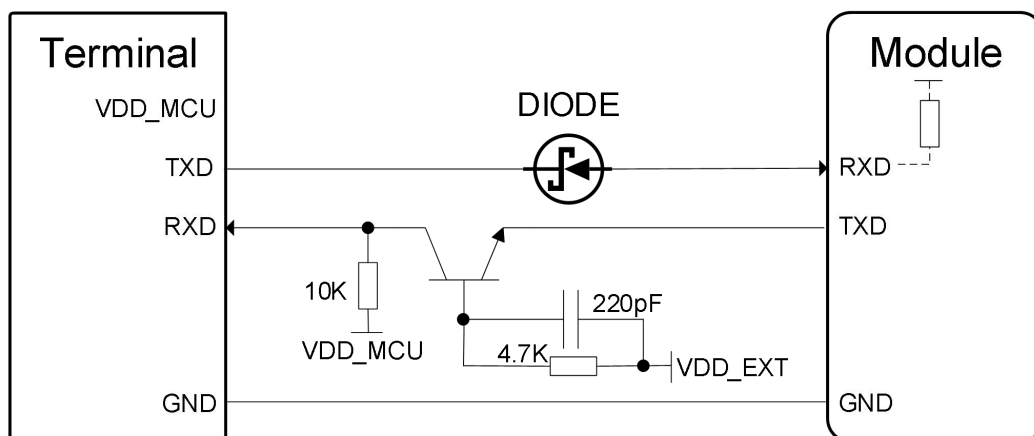


Figure 4.6 Transistor Level Conversion Reference Circuit

Recommended transistors for reference:

Brand: CJ Specification Model: S8050 J3Y Package: SOT-23

Recommended diodes for reference (note that the forward voltage of the diode should be $\leq 0.3V$):

Brand: LRC Specification Model: LRB520S-30T1G Package: SOD-523

4.3 USIM card interface

The module includes 2 external USIM card interfaces. Card 1 complies with ETSI and IMT-2000 standards and supports 1.8 V and 3.0 V USIM cards. Card 2 supports 1.8 V USIM cards by default. The pin description is as follows:

Table 4-3 External USIM Card Interface Pin Definitions

Pin number	Pin name	Description	DC Characteristics
11	USIM_DATA	SIM card data cable	VO_LDOSIM
12	USIM_RST	SIM card reset line	VO_LDOSIM
13	USIM_CLK	SIM card clock line	VO_LDOSIM
15	USIM_VDD	SIM card power supply	Vnorm=1.8/3.0V
79	USIM_DET	SIM Card Status Signal Detection	SIM card slot hot-plug detection interface
62	USIM2_CLK	SIM2 (secondary card) clock line	VO_LDOIO
63	USIM2_RST	SIM2 (secondary card) reset line	Internally connected in parallel with pin 50, cannot be used simultaneously.
64	USIM2_DATA	SIM2 (Secondary SIM) Data Cable	Internally connected in parallel with pin 51, cannot be used simultaneously.
65	USIM2_VDD	SIM2 (secondary card) power supply,	The same power supply as USIM_VDD

The module supports hot-swapping of the USIM card through USIM1_DET and supports both high-level and low-level detection. This feature is disabled by default and can be configured via AT+QSIMDET. If the USIM card detection function is not needed, USIM_DET can be left floating.

4.3.1 USIM Card Circuit Reference Design

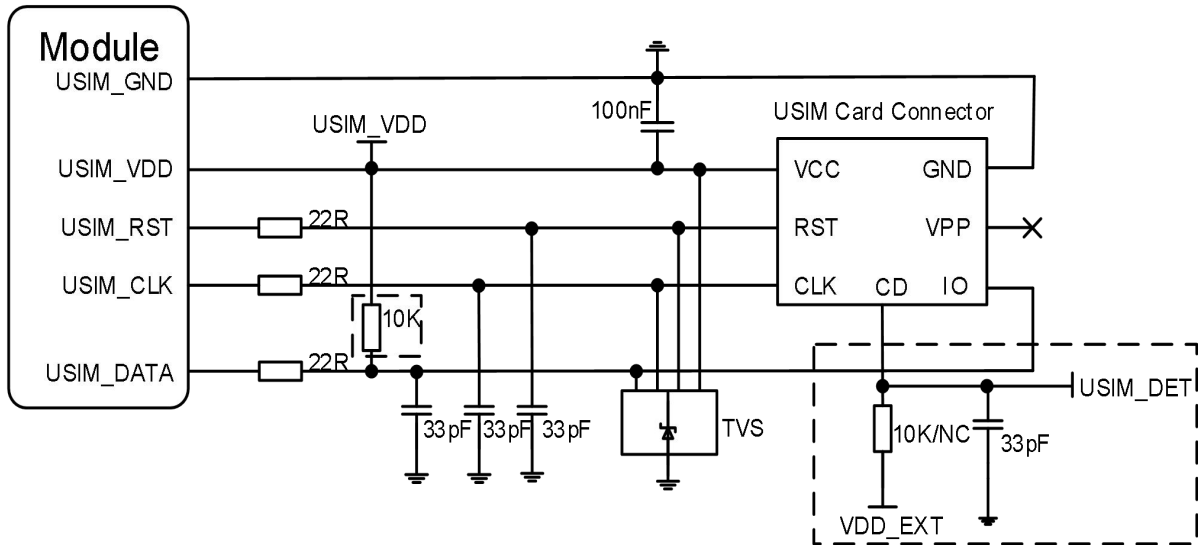


Figure 4.7 6PIN SIM Card Reference Circuit

Precautions

To ensure that the power consumption during sleep mode meets design standards, it is recommended to use a SIM card holder with the CD pin normally closed.

The USIM2 hardware design circuit can also refer to the above design. For details, please refer to the "Lierda NT26-FEU D Series Hardware Reference Design Manual."

4.3.2 Precautions for USIM Card Circuit Design

To ensure the reliability and availability of the SIM card in the application, please follow the following USIM circuit design guidelines:

- ◆ When laying out, try to place the SIM card as close to the module as possible, and keep the signal line wiring length preferably under 200mm;
- ◆ SIM card signal lines should be kept away from RF routing and VBAT power lines;
- ◆ The GND wiring of the SIM card should be short and thick, ensuring the wiring width is not less than 0.5mm;
- ◆ The decoupling capacitor for USIM_VDD should not exceed 1uF, and the capacitor

should be placed close to the VCC of the external SIM card;

◆ To avoid signal crosstalk between USIM_DATA and USIM_CLK, the routing of the two should not be too close, and a ground shield should be added between the two traces. Additionally, to prevent the effects of excessively long traces, USIM_DATA generally requires a pull-up resistor to USIM_VDD to enhance driving capability. If the trace is too long, it is recommended to externally reserve a 10K resistor placed near the card slot. Furthermore, the USIM_RST signal also needs to be shielded with ground protection.

◆ To ensure good ESD protection performance, it is recommended to add a TVS diode to the pins of the external USIM card. The selected TVS diode should have a parasitic capacitance not exceeding 15pF; a 22-ohm resistor should be connected in series on the signal line between the module and the SIM card to suppress stray EMI and enhance ESD protection. At the same time, a parallel 33pF capacitor is used to filter out RF interference. The related resistors, capacitors, and TVS diodes should be placed close to the USIM card socket.

4.4 I2C interface

The module provides one I2C interface, with the interface description as follows:

Table 4-4 I2C Interface Pin Definitions

Pin number	Pin Name	Description	DC Characteristics	Remarks
66	I2C_SDA	I2C data	VO_LDOIO	The I2C data signal requires an external 4.7K pull-up resistor; if not used, it should be left floating.
67	I2C_SCL	I2C clock	VO_LDOIO	The I2C clock signal requires an external 4.7K pull-up resistor; if not used, it should be left floating.

4.5 Simulated audio interface

4.5.1 Simulated audio introduction

Table 4-5 SPK Pin Definitions

Pin number	Pin Name	Description	DC characteristics	Remarks
5	SPK_P	Audio true differential output	VO_LDOIO	Internally connected in parallel with pin 82, cannot be used simultaneously.
6	SPK_N	Audio negative differential output	GND	If not used, leave floating

Remarks

The audio output channel can be used as a speaker (if the output power is insufficient, it is recommended to externally add an audio amplifier device) or as a handset output channel. To enhance anti-interference capability, it is recommended to use differential output.

4.5.2 Simulated Audio Reference Application

High-power speaker playback can be achieved through an external audio PA. A common application reference circuit is as follows:

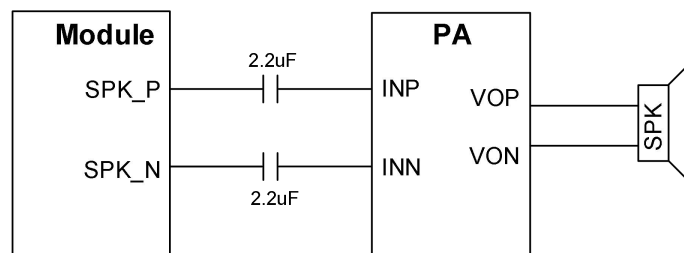


Figure 4.8 SPK Application Reference Circuit

In practical applications, different PAs are chosen, resulting in different circuits. The specific circuit can be found in the document "Lierda NT26KxBx Series Hardware Reference Design Manual."

4.5.3 Audio application notice

The severity of high-frequency interference during GSM transmission usually depends on the application design. In some cases, TDD noise under EGSM900 is more severe, while in other cases, TDD noise under DCS1800 is more severe. Therefore, the required filtering capacitors can be selected based on actual test results. The placement of RF filtering capacitors on the PCB should be as close as possible to the audio components or audio interfaces, and the wiring should be as short as possible, passing through the filtering capacitors first before reaching other points.

To reduce radiation interference, the antenna position should be as far as possible from audio components and audio traces; power traces and audio traces should not be parallel, and power traces should be kept as far away from audio traces as possible.

4.6 ADC interface

The module provides 2 channels of 12-bit analog-to-digital conversion input interfaces (ADC interfaces). When wiring the ADC interfaces, to improve the accuracy of voltage measurement at the interface, it is recommended to implement ground shielding. The pin definitions are as follows:

Table 4-6 ADC Pin Definitions

Pin number	Pin Name	Description	DC Characteristics	Remarks
9	ADC0	General ADC Interface	Input voltage range: 0~1.05V	If not used, leave floating
96	ADC1	General ADC Interface		If not used, leave floating

Using AT+QADC=<port> can read the voltage values of each pin of the ADC interface:

◆ AT+QADC=0 or AT+LADC=0 (please refer to the "AT Command Manual" for specific

commands): can be used to read the voltage value of ADC0.

- ◆ AT+QADC=1 or AT+LADC=1 (for specific commands, please refer to the "AT Command Manual"): can be used to read the voltage value of ADC1.

Table 4-7 ADC Characteristics

Parameters	Minimum value	Typical value	Maximum value	Remarks
ADC voltage range	0	-	1.05	V
ADC input impedance	0.26	-	0.75	Ω
ADC accuracy	-	12	-	bits

Precautions

◆ If the acquisition voltage is greater than 1.05V, it is recommended to use a resistor voltage divider circuit for the ADC pin input. The resistance value of the voltage divider resistors should not exceed 100K Ω , and resistors with an accuracy of more than 1% are recommended; otherwise, the ADC measurement accuracy will be reduced.

◆ Reserve a 1nF capacitor across the grounding voltage divider resistor during design, which can be omitted by default.

4.7 Status indicator signal

The module has one network status indicator and one operating status indicator interface. The interface definitions are as follows:

Table 4-8 Status Indicator Pin Definitions

Pin number	Pin Name	Description	DC Characteristics	Remarks
16	NET_STAT US	Network Status Indicator	VDD18AON	If not used, leave floating
25	STATUS	Operating Status Indicator	LDO_AONIO/VDD18AON	AGPIO function, power domain LDO_AONIO,

				no power loss during sleep. WAKEUP function, power domain VDD18AON, does not lose power during sleep.
--	--	--	--	---

4.7.1 Network Status Indicator

The NET_STATUS pin is used to indicate the network status of the module, and the logic levels for different states are as follows:

Table 4-9 Status Indicator Pin Definitions

Pin Name	Level logic	Network status
NET_STATUS	Slow flash (200ms high / 1800ms low)	Network search status
	Slow flash (1800ms high / 200ms low)	Standby mode
	Flash (125ms high / 125ms low)	Data Transmission Mode

4.7.2 Operating Status Indicator

The STATUS pin is used to indicate the operating status of the module. When the module is powered on normally, STATUS outputs a high level, and it remains effective even in sleep mode.

4.7.3 Indicator Light Circuit Reference Design

Both the NET_STATUS and STATUS pins can be used to drive LED indicators. The reference design is as follows:

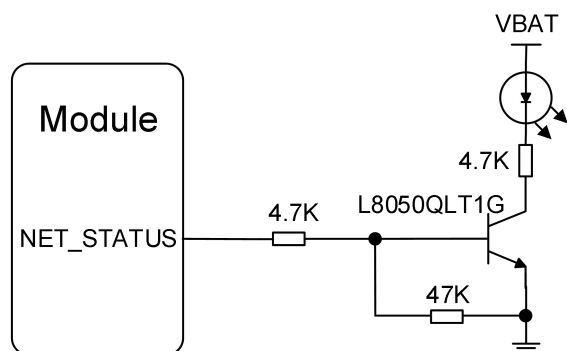


Figure 4-9 Network Status Indicator Light Reference Circuit

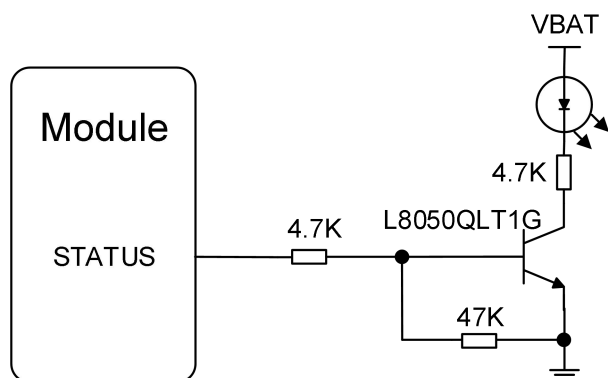


Figure 4-10 Module Status Indicator Light Reference Circuit

5 Radio Frequency Characteristics

5.1 Introduction to RF Functionality

The NT26KxBx series module provides one RF interface with a characteristic impedance of 50Ω . The interface description is as follows:

Table 5-1 RF Interface Pin Definitions

Pin number	Pin Name	I/O	Description	Remarks
35	ANT_MAIN	AIO	Main antenna interface	50Ω characteristic impedance

5.2 Cellular network

5.2.1 Operating frequency band

The NT26KxBx series modules support the following operating frequency bands:

Table 5-2 Operating Frequency Bands

Operating frequency band	Send (MHz)	Receive (MHz)
LTE-FDD B1	1920~1980	2110~2170
LTE-FDD B3	1710~1785	1805~1880
LTE-FDD B5	824~849	869~894
LTE-FDD B8	880~915	925~960
LTE-TDD B34	2010~2025	2010~2025
LTE-TDD B38	2570~2620	2570~2620
LTE-TDD B39	1880~1920	1880~1920
LTE-TDD B40	2300~2400	2300~2400
LTE-TDD B41	2535~2675	2535~2675

Precautions

LTE-TDD B41 only supports 140M (2535-2675MHz).

5.2.2 Transmission power

The transmission power of the NT26KxBx series modules is as follows:

Table 5-3 Conducted Power

Frequency band	Maximum value	Minimum value	Remarks
LTE-FDD B1 (10 MHz)	23dBm±2.7dB	<-40dBm	Compliant with Cat.1 protocols in 3GPP Rel-13 and Rel-14
LTE-FDD B3 (10 MHz)	23dBm±2.7dB	<-40dBm	
LTE-FDD B5 (10 MHz)	23dBm±2.7dB	<-40dBm	
LTE-FDD B8 (10 MHz)	23dBm±2.7dB	<-40dBm	
LTE-TDD B34 (10 MHz)	23dBm±2.7dB	<-40dBm	
LTE-TDD B38 (10 MHz)	23dBm±2.7dB	<-40dBm	
LTE-TDD B39(10 MHz)	23dBm±2.7dB	<-40dBm	
LTE-TDD B40 (10 MHz)	23dBm±2.7dB	<-40dBm	
LTE-TDD B41 (10 MHz)	23dBm±2.7dB	<-40dBm	

5.2.3 Reception sensitivity

The receiving sensitivity of the NT26KxBx series modules is as follows:

Table 5-4 Conduction Reception Sensitivity under Single Transmission (Throughput ≥ 95%)

Frequency band	Reception Sensitivity (10M)	3GPP Standard (10M)
LTE-FDD B1 (10 MHz)	-97.8dBm	-96.3dBm
LTE-FDD B3 (10 MHz)	-99.8Bm	-93.3dBm
LTE-FDD B5 (10 MHz)	-98.3dBm	-94.3dBm
LTE-FDD B8 (10 MHz)	-98.9dBm	-93.3dBm
LTE-TDD B34 (10 MHz)	-101dBm	-96.3dBm
LTE-TDD B38 (10 MHz)	-100.4dBm	-96.3dBm
LTE-TDD B39(10 MHz)	-101.3dBm	-96.3dBm
LTE-TDD B40 (10 MHz)	-100.8dBm	-96.3dBm
LTE-TDD B41 (10 MHz)	-100.3dBm	-94.3dBm

5.3 Antenna Reference Circuit Design

When users are using the NT26KxBx series modules, a π -type matching circuit needs to be added between the module's RF antenna interface and the user's baseboard antenna interface. The typical antenna matching circuit and initial parameters are shown in the figure below:

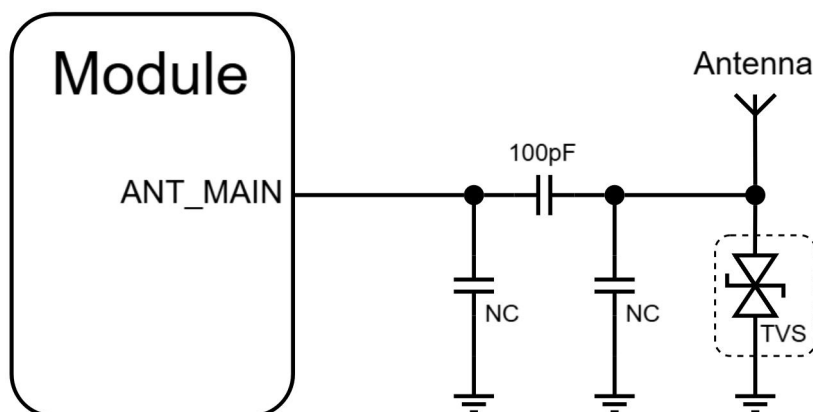


Figure 5.1 RF Antenna Reference Circuit

Precautions

- ◆ The series position uses 100pF, and the capacitor position is not populated by default. The recommended component package is 0201 or 0402.
- ◆ If an external antenna is used, or if the user can touch the antenna, it is recommended to reserve a TVS diode to enhance electrostatic protection. Since the parasitic capacitance of the TVS may affect antenna performance, a model with parasitic capacitance $\leq 0.5\text{pF}$ should be selected. After adding the TVS diode, it is recommended to retune the antenna.

5.4 RF Signal Line Routing Guidelines

The routing between the module antenna interface and the user antenna must comply with the 50-ohm characteristic impedance requirement. At the same time, the RF routing should be as short as possible to ensure that the insertion loss of the RF routing is

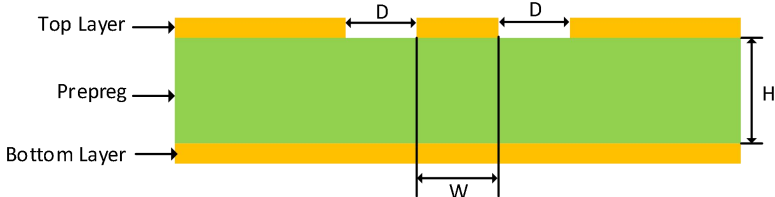
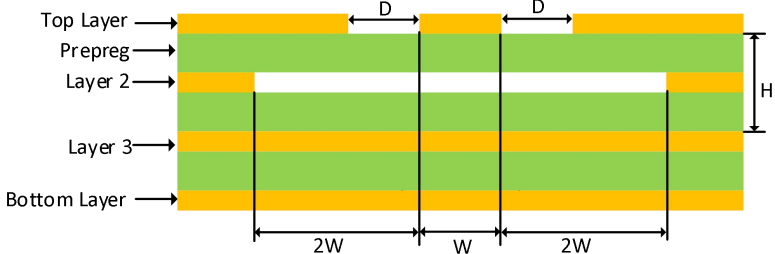
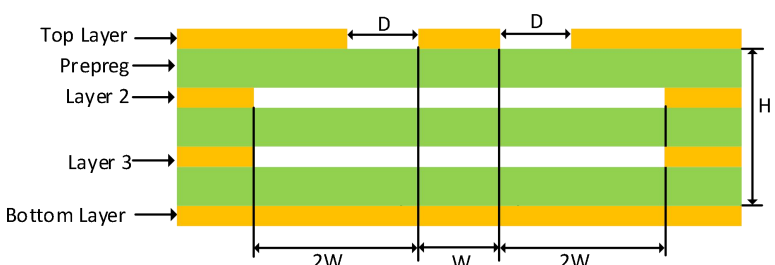
minimized.

Under normal circumstances, the impedance of the RF signal line is determined by the dielectric constant ϵ_r of the material, the trace width W , the distance to ground D , and the thickness H of the reference ground plane.

In the field of Internet of Things applications, the design of PCB characteristic impedance is usually realized using a coplanar waveguide approach, which helps achieve better shielding of RF signal lines while also enabling higher integration for compact area design.

Common PCB coplanar waveguide designs include the following types:

Table 5-5 Design Description of Common PCB Coplanar Waveguide Structures

Serial Number	Pin Name	Remarks
1	 Top Layer → Prepreg → Bottom Layer → Distance D Width W Thickness H	2-layer board The reference ground is the L2 layer.
2	 Top Layer → Prepreg → Layer 2 → Layer 3 → Bottom Layer → Distance D Width W Thickness H 2W	4-layer board The reference ground is the L3 layer. L2 layer hollowing
3	 Top Layer → Prepreg → Layer 2 → Layer 3 → Bottom Layer → Distance D Width W Thickness H 2W	4-layer board The reference layer is L4. L2 and L3 layer hollowing

During the layout process, a 50-ohm impedance can be simulated and calculated using the Polar Si9000 tool. Based on the actual situation, select the appropriate line width, line spacing, and stack-up to achieve the best design results.

Take the calculation example of a 2-layer board with a 1.6mm board thickness PCB

coplanar waveguide:

- ◆ Choose the appropriate PCB substrate material. Once the substrate is selected, the dielectric constant ϵ_r is basically determined (in this example, it is 4.6);
- ◆ Choose an appropriate line width. A good method is to select a line width that matches the size of the pads of the components being soldered (note that it should not be too thin; in this example, 0.65mm is chosen).
- ◆ Based on the selected board thickness, line width, dielectric constant, etc., the spacing to ground can be roughly calculated. If the line spacing is not suitable, the line width can be adjusted to change the line spacing (calculated to be 0.14mm, which is basically appropriate).



Figure 5.2 Reference for the calculation method of 50-ohm impedance

During the actual layout process, the following suggestions are provided for reference:

- ◆ In the π -type circuit, three matched reserved devices are placed close together near the antenna, while bypass devices are recommended to be placed on both sides of the RF line.
- ◆ Place irregular vias on both sides of the RF trace GND plane; there must be a complete GND plane beneath the entire RF trace area.
- ◆ There should be no other wiring beneath the RF line to avoid affecting RF performance or other circuits.

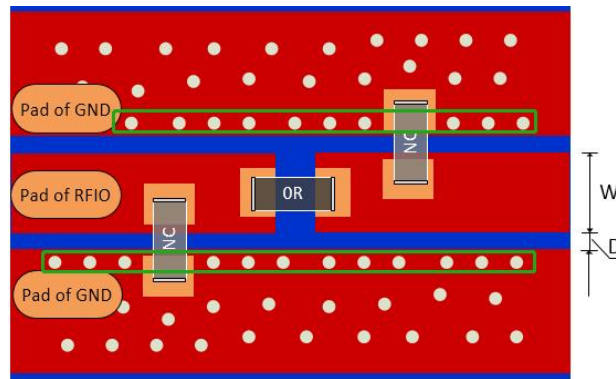


Figure 5-3 RF Routing Layout Design Diagram

The rationality of RF routing can be directly reflected in the module's conduction testing. To ensure the product's overall performance is maximized, antenna design coordination is also required. To better meet the antenna design requirements, the following guidelines are desired in PCB design. Below are instructions for the overall PCB of different layer counts:

◆ When the product PCB is designed with 2 layers, it is best for the TOP and BOTTOM LAYER directly under the module to both be GND layers. The traces that need to be routed from the module should avoid running directly underneath the module and should be routed from the outside of the module.

◆ When the product PCB is designed with 4 layers, it is recommended that the module's routing be placed on the third or fourth layer, reserving the first and second layers as complete GND reference layers for the module.

5.5 Antenna Design Requirements

The RF performance of the module is also affected by the antenna, and the selection of the antenna needs to meet the following requirements:

- ◆ Select an antenna that matches the module's operating frequency band;
- ◆ The antenna's characteristic impedance is required to be 50 ohms to reduce the loss at the connection between the RF line and the antenna;
- ◆ The insertion loss within the operating frequency band should be as low as possible, such as a voltage standing wave ratio (VSWR) ≤ 2 and return loss ≥ 10 dB.

Commonly used antennas suitable for CAT1 scenarios are shown in the following

image:

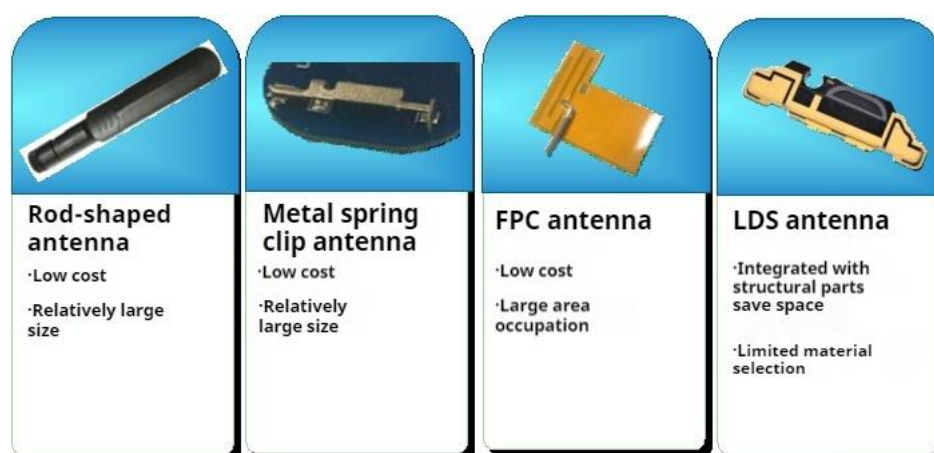


Figure 5.4 Common Antenna Types for CAT1

5.6 Recommended RF connectors

In some application scenarios that require customized antennas, such as when the device is shielded by an iron case, an IPX or SMA connector can be reserved on the motherboard to connect the antenna to the outside of the device via an RF cable, thereby achieving better performance.

Table 5.6 Common RF Connectors

Serial Number	Product Name	Image	Remarks
1	IPX socket		Connect the RF cable
2	SMA connector		Can connect to RF cable or antenna

6 Electrical Performance and Reliability

6.1 Absolute Maximum Ratings

The absolute maximum ratings of the module interface are described as follows:

Table 6-1 Absolute Maximum Ratings

Parameters	Minimum value	Maximum value	Unit
VBAT	-0.3	5.0	V
USB_VBUS voltage	-0.3	5.25	V
Digital interface voltage	-0.3	3.6	V
VBAT current	-	1.2	A

Precautions

When the operating conditions exceed the absolute maximum ratings, permanent damage to the module may occur.

6.2 Power rating

The electrical characteristics of the module power interface are described as follows:

Table 6-2 Electrical Characteristics of Power Interface

Parameters	Description	Conditions	Minimum value	Typical value	Maximum value	Unit
VBAT	Module normal power supply voltage (1)	The actual input voltage must be within this range.	3.3	3.8	4.5	V
VBAT	Module expansion power supply voltage (2)	The actual input voltage must be within this range.	3.1	3.8	4.5	V
IVBAT	Peak current (each) At the time of	At the maximum transmission power level	-	1.0	1.2	A

	transmission slot)					
USB_VBUS	USB detection	-	3.0	5.0	5.25	V
USIM_VDD	USIM card power supply output	-	-	1.8/3.0	-	V
VDD_EXT	Reference power output	-	-	1.8	-	V

Precautions

◆ When the module operates within this voltage range, the module's relevant performance meets the 3GPP standard requirements.

◆ When the module operates within this voltage range, it can still maintain normal operation without irreversible faults; only a few parameters, such as output power, may exceed the 3GPP standard range. When the voltage returns to the normal operating range, all module indicators still comply with the 3GPP standards.

◆ Sleep Mode: VDD_EXT does not power down in sleep mode.

6.3 Power consumption

The power consumption of the module is as follows:

Table 6-3 Module Power Consumption

Module description	Test conditions	Typical value	Unit
Shutdown mode	Module shutdown	0.74	uA
Sleep mode	AT+CFUN=0 (USB disconnected)	6.5	uA
	LTE-FDD @ PF = 32 (USB disconnected)	0.78	mA
	LTE-FDD @ PF = 64 (USB disconnected)	0.41	mA
	LTE-FDD @ PF = 128 (USB disconnected)	0.22	mA
	LTE-FDD @ PF = 256 (USB disconnected)	0.14	mA
	LTE-TDD @ PF = 32 (USB disconnected)	0.75	mA
	LTE-TDD @ PF = 64 (USB disconnected)	0.39	mA

	LTE-TDD @ PF = 128 (USB disconnected)	0.21	mA
	LTE-TDD @ PF = 256 (USB disconnected)	0.15	mA
Idle mode	LTE-FDD @ PF = 64 (USB disconnected)	3.72	mA
	LTE-FDD @ PF = 64 (USB connection)	27.9	mA
	LTE-TDD @ PF = 64 (USB disconnected)	3.5	mA
	LTE-TDD @ PF = 64 (USB connection)	27.6	mA
LTE Data Transmission	LTE-FDD B1	617	mA
	LTE-FDD B3	590	mA
	LTE-FDD B5	592	mA
	LTE-FDD B8	593	mA
	LTE-TDD B34	225	mA
	LTE-TDD B38	222	mA
	LTE-TDD B39	237	mA
	LTE-TDD B40	239	mA
	LTE-TDD B41	223	mA

6.4 Digital Logic Electrical Characteristics

The module digital IO logic level is defined as follows:

Table 6-4 Digital IO Logic Level Description

Type	Parameters	Description	Minimum value	Maximum value	Unit
Input	V _{IL}	Input low level	-0.3	0.6	V
	V _{IH}	Input high level	1.2	2	V
Output	V _{OL}	Output low level	-	0.45	V
	V _{OH}	Output high level	1.35	-	V

Precautions

The module's GPIO reference level follows VDD_EXT, and VDD_EXT powers down during sleep mode, so the corresponding GPIO also powers down. It will lose power.

The AGPIO logic level definitions of the NT26KxBx module are as follows: (AON_VDD range: 1.7~1.9V; typical value: 1.8V)

Table 6-5 Digital IO Logic Level Description

Type	Parameters	Description	Minimum value	Maximum value	Unit
Input	V _{IL}	Input low level	-0.3	0.6	V
	V _{IH}	Input high level	1.2	2	V
Output	V _{OL}	Output low level	-	0.45	V
	V _{OH}	Output high level	1.35	-	V

Precautions

The module's AGPIO reference level follows LDO_AONIO and will not power off during sleep.

The logic level definitions for the module USIM card interface are as follows:

Table 6-6 USIM 1.8V Logic Level Description

Type	Parameters	Description	Minimum value	Maximum value	Unit
Input	V _{IL}	Input low level	-	0.6	V
	V _{IH}	Input high level	1.2	-	V
Output	V _{OL}	Output low level	-	0.45	V
	V _{OH}	Output high level	1.35	-	V

Table 6-7 USIM 3.0V Logic Level Description

Type	Parameters	Description	Minimum value	Maximum value	Unit
Input	V _{IL}	Input low level	-	1.0	V
	V _{IH}	Input high level	1.95	-	V
Output	V _{OL}	Output low level	-	0.45	V
	V _{OH}	Output high level	2.55	-	V

Precautions

The USIM_VDD level is determined based on the detected type of USIM card and supports 1.8V/3.0V USIM cards.

6.5 Electrostatic protection

In daily life and production, static electricity is everywhere. Static electricity generated by the human body or by friction between objects can be transmitted to modules through various means and may potentially damage the modules. Therefore, it is essential to pay close attention to static protection and take necessary anti-static measures. For example:

- ◆ Wear anti-static gloves during research and development, production, assembly, and testing processes;
- ◆ When designing products, add anti-static protection devices at circuit interfaces and other points susceptible to electrostatic discharge.

The electrostatic discharge performance of the module is as follows:

Table 6-8 Electrostatic Protection Characteristics

Test interface	Contact discharge	Remarks
VBAT	$\pm 8\text{KV}$	Test Standard: IEC61000-4-2 Temperature: 25°C Humidity: 45%
GND	$\pm 8\text{KV}$	
Antenna interface	$\pm 8\text{KV}$	
Others	-	

For detailed electrostatic protection design, refer to the "Lierda NT26KxBx Series Hardware Reference Design Manual."

6.6 Operating and storage temperature

Table 6-9 Operating Temperature Range

Parameters	Minimum value	Typical value	Maximum value	Unit	Remarks
Normal operating temperature (1)	-35	+25	+75	°C	
Extended Operating Temperature (2)	-40	+25	+85	°C	
Storage ambient temperature(3)	-40	+25	+90	°C	

Precautions

(1) When the module operates within this temperature range, its related performance meets the 3GPP standard requirements.

(2) When the module operates within this temperature range, it can still maintain normal operation without any unrecoverable failures; only a few parameters, such as output power, may exceed the 3GPP standard range. When the temperature returns to the normal operating range, all module parameters will still comply with the 3GPP standards.

(3) This storage temperature range does not include the packaging materials; attention should be paid to the maximum tolerance temperature of the tape packaging.

6.7 Precautions

There are some precautions to be noted during the production and use of the module:

- ◆ When spraying the module, please try to avoid spraying material into the inside of the module;
- ◆ When cleaning the module, do not use ultrasonic cleaning on the module, as it may cause damage to the internal crystal of the module.
- ◆ When producing and using the module, please avoid applying it in environments or packages containing any amount of mercury or mercury vapor, as this may lead to product failure or malfunction risk.

7 Mechanical dimensions

7.1 Mechanical Dimensions

The module mechanical dimension diagram is as follows:

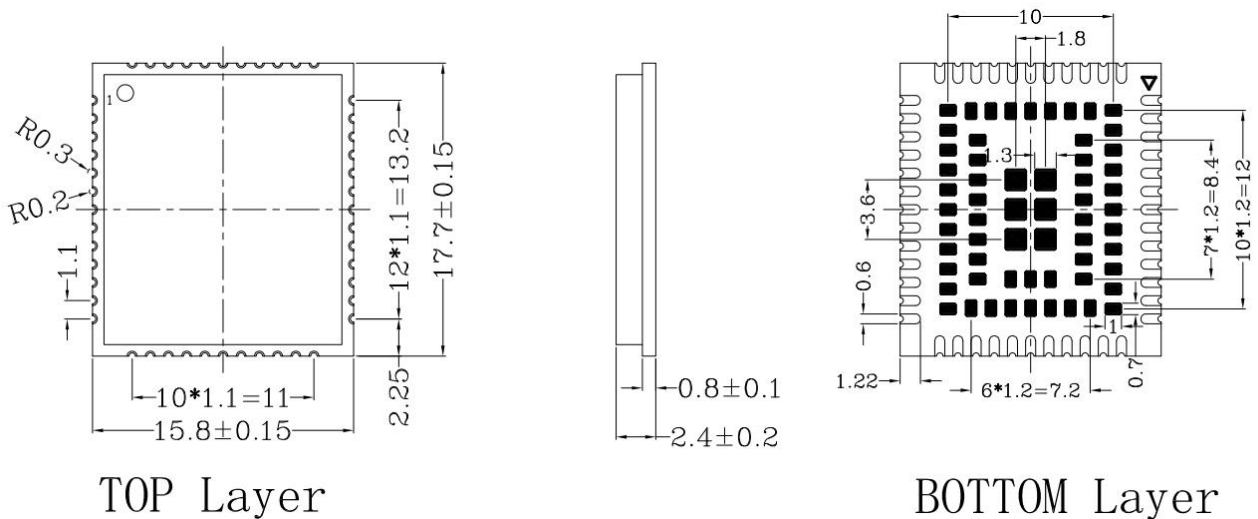


Figure 7.1 Module Mechanical Dimensions (Unit: mm)

7.2 Module Top View/Bottom View

The actual module effect diagram is as follows, subject to the actual module.



Figure 7.2 Module Top/Bottom View

7.3 Recommended packaging

The recommended module soldering pads are shown in the figure below. Users can

◆ The module's pins around the perimeter feature rounded corners internally. When designing the PCB pads, users should consider using rounded corner transitions. For the rectangular pads at the bottom of the module, the PCB design can use the module pin dimensions, as shown in the reference design diagram of a single pad below.

◆ To facilitate the opening of step stencils, it is recommended not to place other components within 2.0mm outside the module pads. Users can refer to their own stencil supplier's requirements to determine this distance.

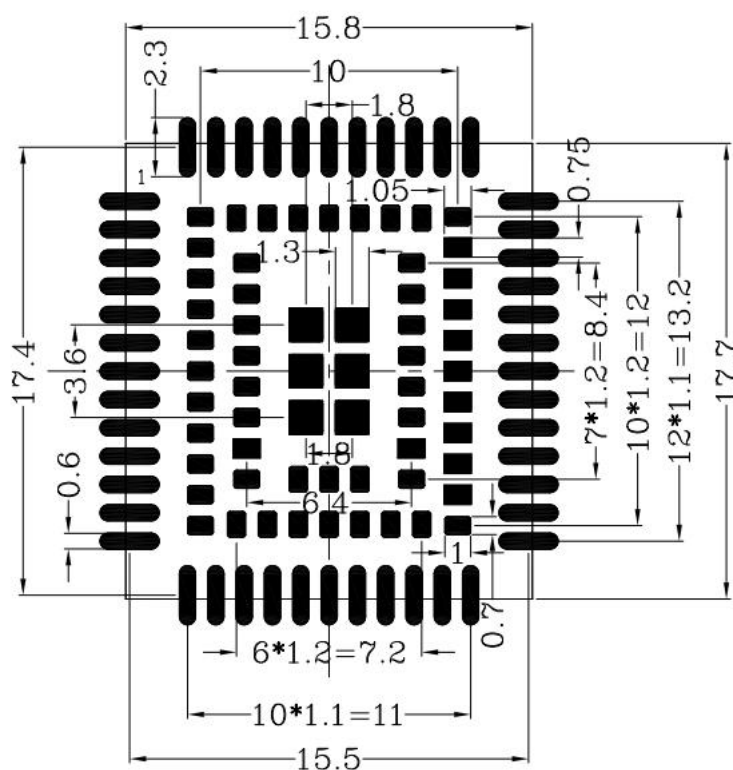


Figure 7.3 Recommended Module Pad (Unit: mm)

8 Production and Packaging Information

This chapter describes the module's surface-mount technology, packaging methods, storage conditions, and other information, which can help users better preserve and use the module.

8.1 Production welding

8.1.1 Furnace processing method

When mounting the module, the following precautions should be observed during the reflow process:

- ◆ To ensure the quality of module welding, the recommended thickness of the steel mesh corresponding to the module pad section is 0.18-0.2mm;
- ◆ If the module's baseboard is double-sided SMT, it is recommended to place the module during the second SMT process to avoid damage to the module caused by repeated heating.
- ◆ During the first chip mounting, it is best for the customer's substrate to pass through the oven on the conveyor belt; for the second chip mounting, it is also preferable to place it on the conveyor belt for oven passing.
- ◆ If for special reasons the PCBAs cannot be placed on the conveyor belt for reflow, consider using a fixture to pass through the reflow oven on the track or placing a flat, high-temperature resistant, straight template to support the PCBA during reflow, preventing PCB deformation that could cause poor solder joints in the module.

8.1.2 Reflow Soldering Operation Instructions

The PCBA reflow oven temperature profile is related to the solder paste used and needs to be adjusted according to the actual solder paste. The data is only suitable for lead-free operations. Refer to Figure 8-2 Lead-Free Reflow Soldering Operation Guide.

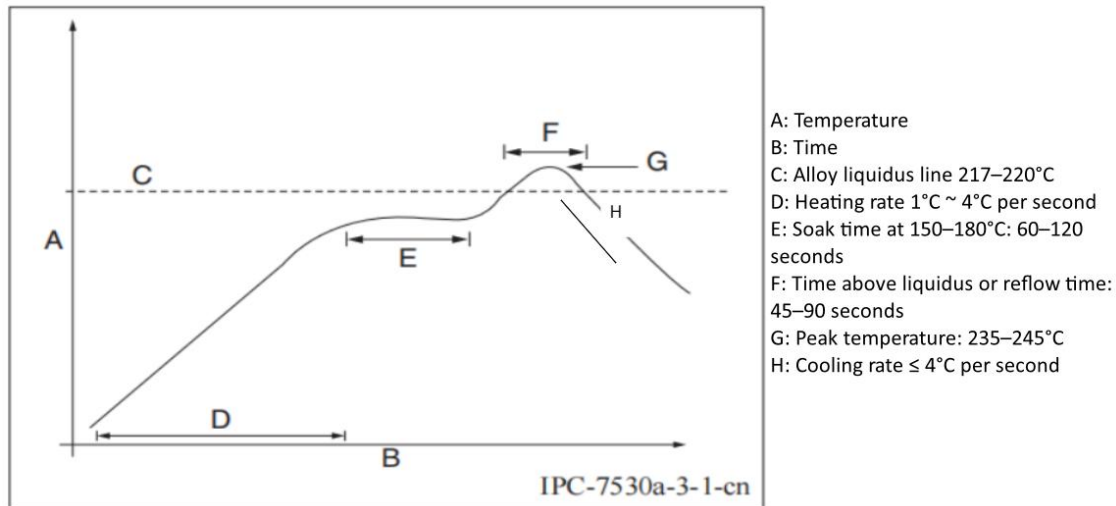


Figure 8.1 Lead-free Reflow Soldering Operation Guide

Remarks

(1) During the production welding or other processes that may directly contact the module, no organic solvents (such as alcohol, isopropanol, acetone, trichloroethylene, etc.) shall be used to wipe the module shield; otherwise, it may cause the shield to rust.

(2) The modular nickel silver laser-engraved shielding cover meets the following requirements: after 12 hours of neutral salt spray testing, the laser-engraved information remains clear and identifiable, and the QR code can be scanned (there may be white corrosion).

(3) If the module needs to be sprayed, please ensure that the spraying material used does not chemically react with the module shield or PCB, and also ensure that the spraying material does not flow into the inside of the module.

(4) Please do not ultrasonic clean the module, as it may cause damage to the internal crystal of the module.

(5) Due to the complexity of the SMT process, if you encounter uncertain situations or processes not mentioned in the documentation (such as selective wave soldering or ultrasonic welding), please confirm with Lierda technical support before starting the SMT process.

8.1.3 Maintenance

If the module has defects such as cold solder joints or short circuits and requires repair, please follow the parameters below:

◆ Lead-free process: Soldering iron temperature $380 \pm 10^{\circ}\text{C}$, soldering iron contact time ≤ 5 seconds.

◆ Leaded process: Soldering iron temperature $350 \pm 10^{\circ}\text{C}$, soldering iron contact time ≤ 5 seconds.

8.2 Packaging specifications

The module's factory packaging uses a rubber wheel tape method, with the rubber wheel reference dimensions as follows:

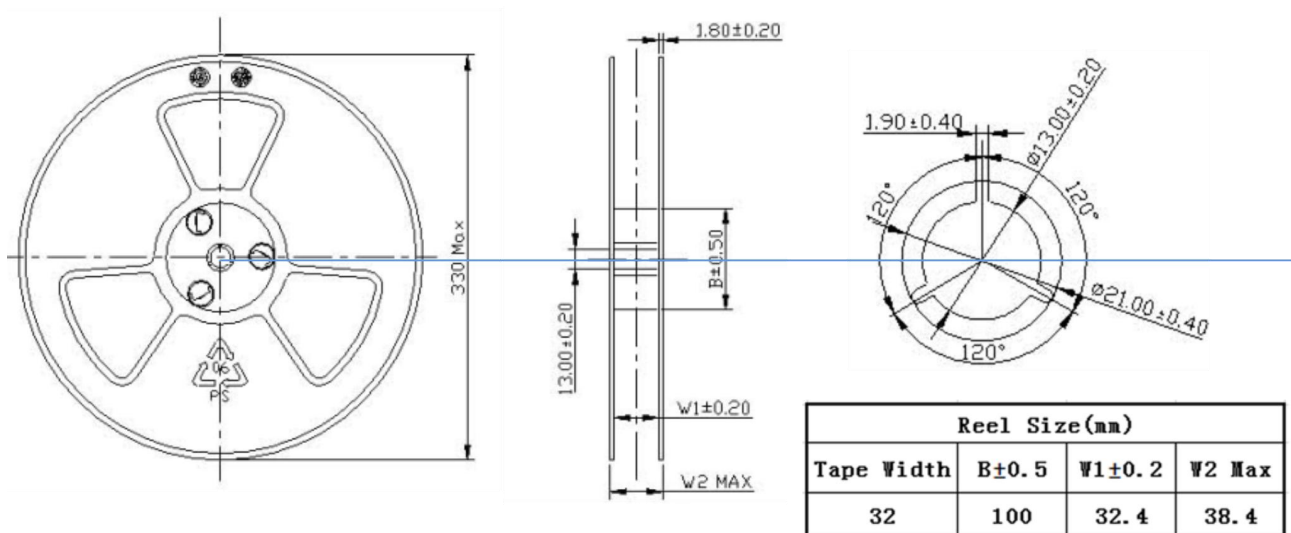


Figure 8.2 Carrier Belt Roller Dimension Diagram

The tape feeding direction is as follows: (Refer to the diagram, the label content is subject to the actual product, and pay attention to the module PIN1 position.)

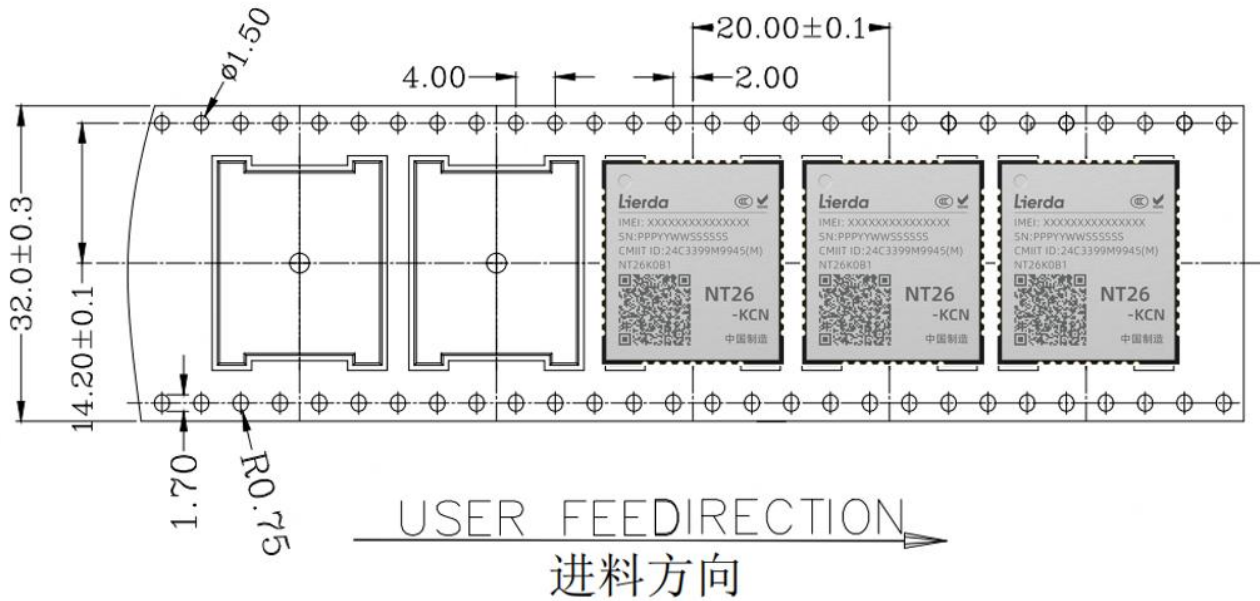


Figure 8.3 Carrier Feed Direction Indicator Diagram

8.3 Storage conditions

The module is shipped in the form of a vacuum-sealed bag. The module's moisture sensitivity level is 3 (MSL 3), and its storage must comply with the following conditions:

1. Recommended storage conditions: temperature 23 ± 5 °C, and relative humidity 35~60%.
2. Under the recommended storage conditions, the module can be stored in a vacuum-sealed bag for 12 months.
3. Under workshop conditions of 23 ± 5 °C and relative humidity below 60%, the shelf life of the module after unpacking is 168 hours. Under these conditions, the module can be directly subjected to reflow production or other high-temperature operations. Otherwise, the module needs to be stored in an environment with relative humidity below 10% (e.g., a moisture-proof cabinet) to keep the module dry.
4. If the module is under the following conditions, pre-baking treatment is required to prevent the module from absorbing moisture and causing PCB blistering, cracking, and delamination after high-temperature soldering:

- Storage temperature and humidity do not meet the recommended storage conditions;

- After the module is unsealed, production or storage is not completed according to Article 3 above;

- Vacuum packaging leaks, material bulk packaging;
- Before module repair.

5. Module baking process:

- Requires high-temperature baking at 120 ± 5 °C for 8 hours;
- Modules that undergo a second baking must be soldered within 24 hours after baking; otherwise, they need to be stored in a drying oven.

Precautions

(1) To prevent and reduce welding defects such as blistering and delamination caused by moisture absorption in the module, strict control must be implemented, and the vacuum package should not be opened and exposed to the air for an extended period.

(2) Before baking, remove the original packaging and transfer the module to a high-temperature resistant carrier to prevent the plastic tray or reel from deforming due to heat. If only a short baking time is required, refer to the IPC/JEDEC J-STD-033 standard.

(3) When unpacking and transferring modules, anti-static measures must be taken, such as wearing anti-static gloves.

9 Relevant Documents and Terminology

Abbreviations

9.1 Related documents

The relevant document references are as follows:

Table 9-1 Related Documents

Serial Number	Document Name
[1]	Lierda NT26KxBx Series Hardware Reference Design Manual
[2]	Lierda LTE Standard (L) Series AT Command Manual
[3]	Lierda LTE Standard(Q) Series AT Command Manual
[4]	IPC/JEDEC J-STD-033 Standard

9.2 Terminology Abbreviation

The abbreviations of terms involved in this document and their related definitions are as follows:

Table 9-2 Terminology Abbreviations

Abbreviation	Full English name	Full Chinese Name
3GPP	3rd Generation Partnership Project	Third Generation Partner Program
ADC	Analog-to-Digital Converter	Analog-to-digital conversion
ANT	Antenna	Antenna
ASM	Antenna Switch Module	Antenna switch module
DAC	Digital -to- Analog Converter	Analog-to-digital conversion
DBG	Debug	Debugging
DC-DC	Direct Current - Direct Current	DC converter
DCXO	Digitally Controlled Crystal Oscillator	Digital Control Crystal Oscillator
DRX	Discontinuous Reception	Non-continuous reception
DTE	Data Terminal Equipment	Data terminal equipment

ECL	Equivalent Class Level	Network Coverage Level
ESD	Electro-Static discharge	Electrostatic discharge
EOS	Electrical Overstress	Electrical Overstress (Surge)
ESR	Equivalent Series Resistance	Equivalent series resistance
EVK	Evaluation Kit	Evaluation Toolkit
H-FDD	Half Frequency Division Duplexing	Frequency Division Half Duplex
FOTA	Firmware Over-The-Air	Remote firmware upgrade
GPIO	General-purpose input/output	General input and output
I/O	Input/Output	Input and Output Interface
I _{max}	Maximum Load Current	Maximum current
I _{norm}	Normal Current	Normal (typical) current
bps	Bits Per Second	Rate unit
LCC	Leadless Chip Carriers	Leadless chip carrier packaging
LCM	LCD Module	LCD display module
LDO	Low Dropout Regulator	Low dropout linear regulator
LGA	Land Grid Array	Grid array packaging
LwM2M	Lightweight Machine-To-Machine	Lightweight M2M Protocol
MCU	Micro Controller Unit	Microcontroller Unit
MSL	Moisture Sensitivity levels	Moisture Sensitivity Level
PCB	Printed Circuit Board	Printed Circuit Board
PCBA	Printed Circuit Board Assembly	Printed Circuit Board Assembly
PMU	Power Management Unit	Power Management Unit
PSM	Power Saving Mode	Energy-saving mode
RF	Radio Frequency	Radio frequency
RoHS	Restriction of Hazardous Substances	Restriction of hazardous substances

RX	Receive	Received
TAU	Tracking Area Update	Tracking area update
TCP/IP	Transmission Control Protocol/Internet Protocol	Transmission Control Protocol/Internet Protocol
TVS	Transient Voltage Suppressor	Transient suppression diode
TX	Transmit	Send
UART	Universal Asynchronous Receiver & Transmitter	General Asynchronous Receiver and Transmitter
UDP/IP	User Datagram Protocol - Internet Protocol	User Datagram Protocol
URC	Unsolicited Result Code	Non-request result code
(U)SIM	(Universal) Subscriber Identification Module	(General) User Identity Recognition Module
VSWR	Voltage Standing Wave Ratio	Voltage Standing Wave Ratio
Vmax	Maximum Voltage Value	Maximum voltage
Vnorm	Normal Voltage Value	Normal (typical) voltage
Vmin	Minimum Voltage Value	Minimum voltage
VIHmax	Maximum Input High Level Voltage Value	Maximum input high level
VIHmin	Minimum Input High Level Voltage Value	Minimum input high level
VILmax	Maximum Input Low Level Voltage Value	Maximum input low level
VILmin	Minimum Input Low Level Voltage Value	Minimum input low level
VImax	Absolute Maximum Input Voltage Value	Maximum input level
VImin	Absolute Minimum Input Voltage Value	Minimum input level
VOHmax	Maximum Output High Level Voltage Value	Maximum output high level
VOHmin	Minimum Output High Level Voltage Value	Minimum output high level
VOLmax	Maximum Output Low Level Voltage Value	Maximum output low level
VOLmin	Minimum Output Low Level Voltage Value	Minimum output low level